

REVIEW

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Lithium-ion batteries: outlook on present, future, and hybridized technologies

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Lithium-ion batteries (LIBs) continue to draw vast attention as a promising energy storage technology due to their high energy density, low self-discharge property, nearly zero-memory effect, high open circuit voltage, and long lifespan. In particular, high-energy density lithium-ion batteries are considered as the ideal power source for electric vehicles (EVs) and hybrid electric vehicles (HEVs) in the automotive industry, in recent years. This review discusses key aspects of the present and the future battery technologies on the basis of the working electrode. We then discuss how lithium-ion batteries evolve to meet the growing demand on high charge capacity and electrode stability. An account of a stand-alone energy device (off-grid system) that combines an energy harvesting technology with a lithium-ion battery is also provided. The main discussion is categorized into three perspectives such as the evolution from the conventional to the advanced LIBs (e.g., Li-rich transition metal oxide and Ni-rich transition metal oxide batteries), to the state-of-the-art LIBs (e.g., Li-air, Li-sulfur batteries, organic electrode batteries, solid-state batteries, and Li-CO₂ batteries), and to the hybridized LIBs (e.g., metal halide perovskite batteries).

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1. Introduction

Lithium-ion batteries (LIBs) are extensively employed as power sources in a wide range of applications owing to their high energy densities, high coulombic efficiencies, low self-discharge features, and a range of chemical potentials accessible with diverse electrode designs.^{1–4} Especially, the past decade has seen a growing trend towards next-generation lithium-ion batteries with high charge capacities/power densities developed for electric vehicles (EVs), hybrid electric vehicles (HEVs), aerospace applications, and autonomous electric devices (e.g. hybrid solar batteries).^{2,5–9} However, conventional lithium-ion batteries, which are mainly based on the Li⁺ intercalation mechanism (e.g. LiCoO₂, LiFePO₄, and LiMn₂O₄), cannot offer the high-charge capacities (>200 mA h g^{−1}) required for particular applications.^{10,11} It is therefore that researchers have shown immense interest recently in the high-energy density battery technologies such as Li-rich (Li_{1.2}Mn_{0.54}Ni_{0.13}Co_{0.13}O₂) and Ni-rich transition metal oxides (e.g. LiNi_{0.8}Co_{0.15}Al_{0.05}O₂ or Li[Ni_{0.8}Co_{0.1}Mn_{0.1}]O₂), high voltage spinels, and Li-sulfur, Li-air, organic electrode, Li-CO₂, and solid-state batteries. They benefit not only from the high-charge capacity, but also from the lower cost. Also, they are less toxic as compared with the widely commercialized LIBs (e.g. LiCoO₂). There are, however,

crucial issues, which need to be addressed for their successful commercialization, such as phase transition, electrolyte decomposition, formation of an unstable solid-electrolyte interphase (SEI), volume expansion and/or surface reconstruction of active material particles, and electrode degradation during charge and discharge.^{2,3} On the other side, a hybrid device that merges two different energy applications (e.g. a solar-cell and a lithium-ion battery) has drawn much attention in the most recent years. This new technology, often regarded as a stand-alone energy system, provides high energy efficiency with reliable power in a remote location.^{7–9} For instance, a photo-rechargeable perovskite ((C₆H₉C₂H₄NH₃)₂PbX₄, X = I or Br) battery can both convert and store energy, serving as an off-grid system.¹² However, an alternative cell configuration with a new class of electrode material is required to realize these types of applications.

In this review, we will discuss the key issues of the present and commercialized lithium-ion batteries, on the basis of the working electrodes. A further discussion will be focused on some state-of-the-art battery technologies, comparing their advantages and disadvantages. Also, particular attention is devoted to a hybridized device (off-grid system) that combines an energy harvesting application with the energy storage technology. Taken together, we present the evolution of lithium-ion batteries (working electrodes) from conventional technologies into advanced, state-of-the-art, and hybridized technologies. The current survey sheds some light on understanding the future direction of the next-generation lithium-ion batteries, Fig. 1.

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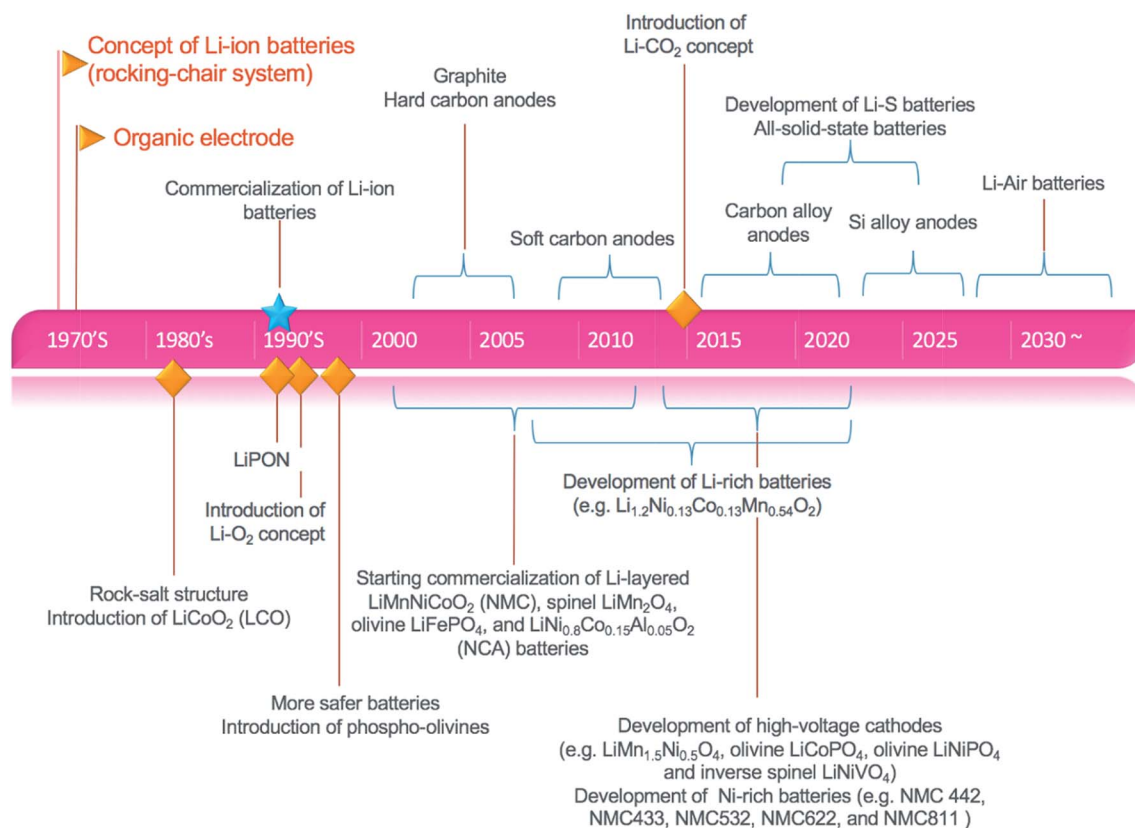


Fig. 1 Historical evolution and advances of lithium-ion battery technologies.

2. Conventional and present technologies

2.1 Single transition metal oxide batteries

The concept of lithium-ion battery was first demonstrated in the early 1970s by Whittingham employing metal dichalcogenides (e.g. TiS_2) as an intercalation compound.^{13,14} The layered structure of TiS_2 comprising Ti and S sheets in a close-packed arrangement allows the intercalation of Li^+ into the host material. This was further investigated by Goodenough, suggesting a design consideration of the electrodes for lithium-ion batteries.¹⁵ Mizushima and Goodenough *et al.* introduced $\text{Li}_x\text{-CoO}_2$ (LCO) with an ordered rock-salt structure as a new cathode material for batteries in the early 1980s.¹⁶ Since then, LCO has been vastly used due to the successful commercialization of the LiCoO_2/C cell by Sony in 1991.¹⁷ The LCO cathode benefits from high energy density, good electrical conductivity, high open-circuit voltage (~ 4.0 V), and low self-discharge.^{16,18,19} The octahedral array of the edge-shared CoO_6 is the key for the good electronic conductivity. Its high energy density is attributed to the two-dimensional migration channel of Li^+ , which is a layered structure of three CoO_2 layers per unit cell with a stacking order of ABCABC. LCO has a theoretical capacity of 274 mA h g^{-1} , when one mole of Li^+ is deintercalated. However, only 140 mA h g^{-1} , which is half of the theoretical capacity, can be utilized practically because of its physical/chemical instabilities stemming from band overlapping between $\text{Co}^{3+/4+}:\text{t}_{2g}$

and $\text{O}^{2-}:\text{2p}$ in the energy diagram (Fig. 3a). In a high voltage range, the oxidation state changes from Co^{3+} into Co^{4+} , followed by the removal of electrons from the $\text{O}^{2-}:\text{2p}$ band, which in turn causes some phase transitions in the crystal structure.²⁰ Several attempts have been made to replace Co with different transition metal elements that minimize the overlap with the $\text{O}^{2-}:\text{2p}$ band, and offer ecological and economical advantages (e.g. Ni and Mn). LiNiO_2 (LNO) is known to be isostructural with LCO, which is generally denoted as the O3-structure based on Delmas notation.²¹ As implied in the energy diagram (Fig. 3a), LNO only involves the e_g band, which lies well above the $\text{O}^{2-}:\text{2p}$ band, during the Li^+ deintercalation. This is a major benefit of using LNO, as it doesn't undergo changes in the lattice oxygen even at a deep deintercalation level of Li^+ .^{22–24} In spite of these advantages, LNO is not a promising cathode material as it suffers from critical drawbacks. It is challenging to synthesize a stoichiometric LNO with pure Ni^{3+} element at high temperatures, because of the instability of Ni^{3+} above 250°C .²⁵ The synthesis process to make LNO with a layered $\alpha\text{-NaFeO}_2$ structure normally employs $\text{Ni}^{2+}/\text{Li}^+$ -based precursors (e.g. LiOH-Ni(OH)_2) at a high temperature ($\sim 700^\circ\text{C}$).^{19,26} As a consequence, the presence of Ni^{2+} in the lithium 3a-site (trigonal, $R\bar{3}m$) is unavoidable after the synthesis, resulting in poor rate capability by impeding Li^+ diffusion.²⁷ It has also been found that LNO undergoes a series of phase transitions at different intercalation states.^{26,28,29} The cathode can be electrochemically activated when it exits as a trigonal structure. However, when $0.5 < x <$

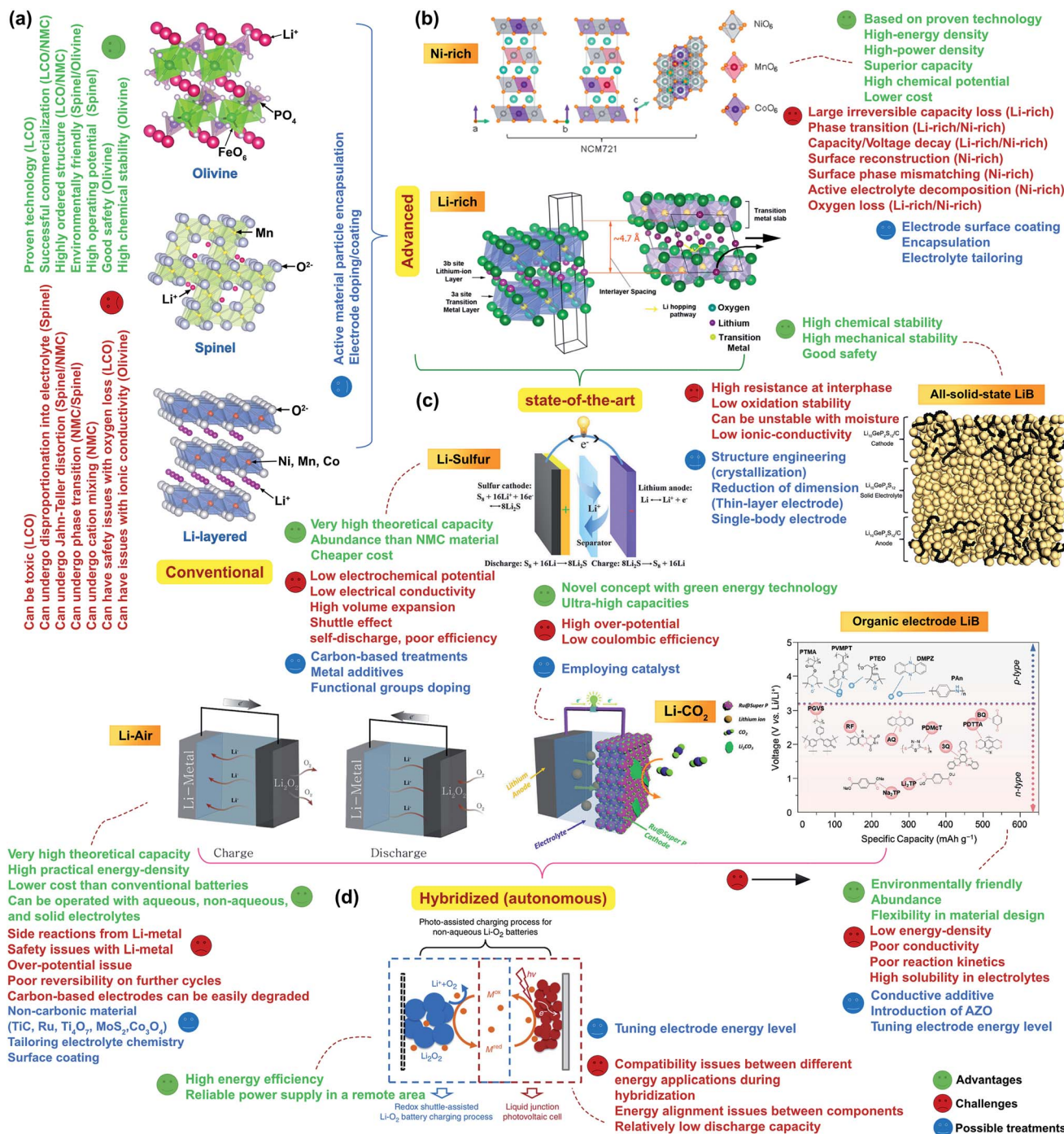


Fig. 2 Evolution of working electrodes for lithium-ion batteries. (a) Conventional systems: olivine, spinel, and Li-layered transition metal oxide cathodes. (b) Advanced systems from conventional systems. Crystal structures of Ni-rich transition metal oxide cathodes and Li-rich transition metal oxide cathodes with an integrated structure between Li_2MnO_3 (monoclinic, $C2/m$) and LiTO_2 (trigonal, $R\bar{3}m$), where T represents Mn, Co, and Ni. Reproduced from ref. 200 with permission (Copyright 2015, American Chemical Society) and reproduced from ref. 201 with permission (Copyright 2016, Elsevier), respectively. (c) Comparison of the working electrodes of Li-sulfur: reproduced from ref. 202 with permission (Copyright 2016, Royal Society of Chemistry), all-solid-state LIBs: reproduced from ref. 203 with permission (Copyright 2015, WILEY-VCH), Li-air, Li-CO₂: reproduced from ref. 159 with permission (Copyright 2017, Royal Society of Chemistry), and organic electrode LIBs: reproduced from ref. 119 with permission (Copyright 2018, WILEY-VCH). (d) Photo-assisted charging process with a photoelectrode incorporated into a lithium-oxygen battery. Reproduced from ref. 204 with permission (Copyright 2014, Nature Publishing Group).

0.75, Li_xNiO_2 transforms into a monoclinic phase.^{26,29} Such phase transitions in the crystal structure can lead to severe capacity fading of electrochemical cells. Jahn-Teller distortion

resulting from the low-spin $\text{Ni}^{3+}:d^7(t_{2g}^6e_g^1)$ is another major problem (Fig. 3b). This was first reported by Rougier *et al.*, revealing the appearance of a NiO_6 distortion in LNO.³⁰ Two

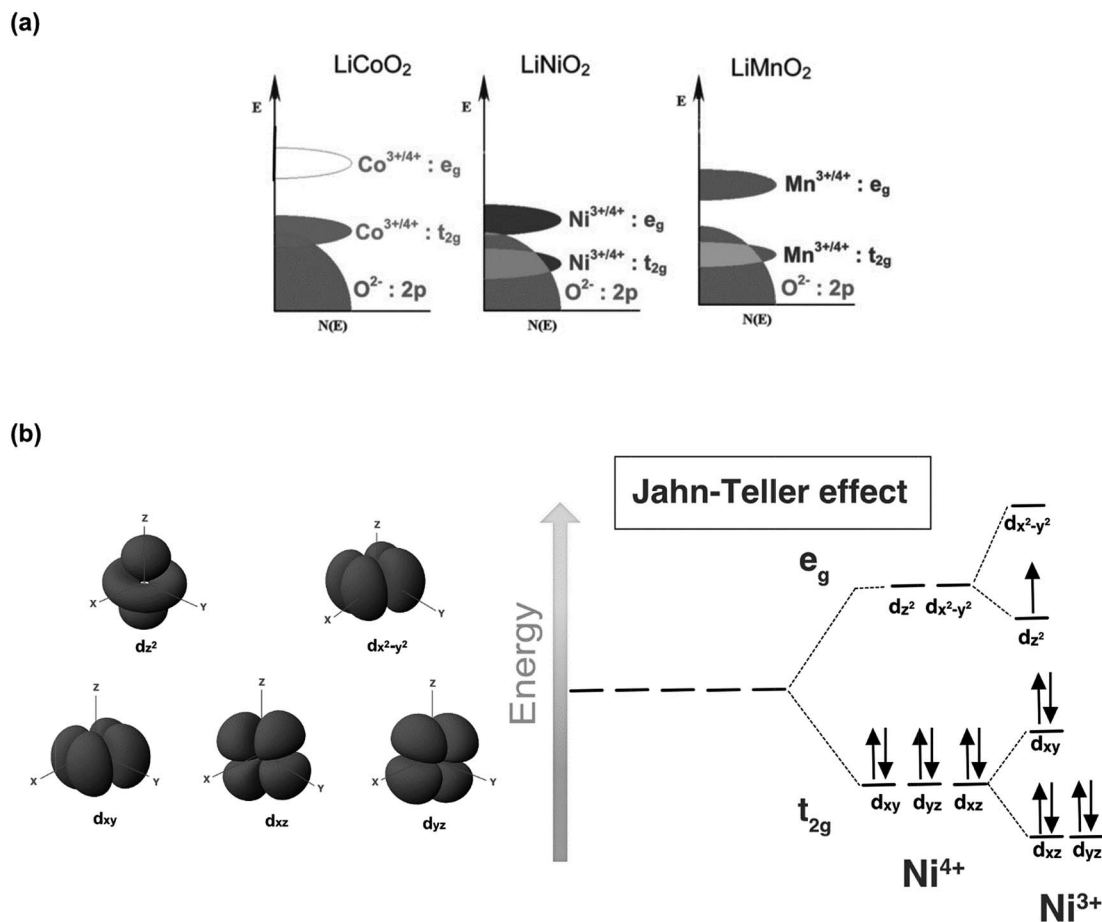


Fig. 3 (a) The energy vs. density of states with the relative Fermi level (E_F) and redox couples of $\text{Co}^{4+/3+}$, $\text{Ni}^{4+/3+}$, and $\text{Mn}^{4+/3+}$ for LiCoO_2 , LiNiO_2 , and LiMnO_2 , respectively. Reproduced from ref. 205 with permission (Copyright 2008, Royal Society of Chemistry). (b) Illustration of the Jahn-Teller effect of NiO_6 with energy splitting in an octahedral complex.

types of different bond lengths (four 1.91 Å and two 2.09 Å bonds) in NiO_6 caused the distortion of the octahedron structure of LNO. LiMnO_2 has also been considered as an alternative electrode material, which can replace LCO due to its high theoretical capacity.^{31,32} LiMnO_2 offers a high capacity of $\sim 285 \text{ mA h g}^{-1}$ and possesses higher power density.^{31,33} Moreover, LiMnO_2 is inexpensive, and it is environmentally benign compared to LCO and LNO. Although the chemical formula of LiMnO_2 is similar to that of LCO and LNO, the structure is not identical to the layered O3-structure as it crystallizes in an orthorhombic phase (space group: $Pmmm$).³⁴ LiMnO_2 with a layered structure is not thermodynamically stable. An ion-exchange method between Na^+ and Li^+ in layered NaMnO_2 has been shown to yield a layered structure of LiMnO_2 .³² It has been reported that LiMnO_2 undergoes a phase transition from the layered structure to a spinel structure (LiMn_2O_4), resulting in a poor cycle life on charge and discharge.³⁵

2.2 Mixed transition metal oxide batteries

The last decade has seen a growing interest in mixed transition metal cathodes $\text{Li}(\text{Ni}_x\text{Mn}_y\text{Co}_{1-x-y})\text{O}_2$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq x + y \leq 1$) owing to their synergetic benefits over a single

transition metal. A mixed transition metal cathode that combines Mn, Co, and Ni with a layered structure of $\alpha\text{-NaFeO}_2$ -type is often denoted as an NMC cathode. It benefits from the high charge capacity of LiNiO_2 , the rate capability of LiCoO_2 , and the structural stability due to Mn^{4+} .^{36,37} NMC delivers a reversible capacity of 160 mA h g^{-1} in the voltage range between 2.5 V and 4.4 V, and can offer 200 mA h g^{-1} when the cell is charged between 2.8 V and 4.6 V.³⁸ NMC is also cheaper than LCO, as the proportion of Co is reduced by Ni and Mn. On the one hand, the presence of Co^{3+} appears to be useful in suppressing the structural distortion stemming from the Ni^{3+} Jahn-Teller effect by disrupting the $\text{Ni}^{2+}/\text{Mn}^{4+}$ cation ordering to some extent.^{39–41} However, in spite of the synergetic merits between the transition metals, NMC batteries can undergo surface reconstruction or unstable SEI formation on the electrode with transition metal decomposition.^{42–44}

2.3 Spinel and olivine cathodes

Spinel LiMn_2O_4 crystallizes in the formula of $\text{A}[\text{B}_2]\text{O}_4$ (cubic, $Fd\bar{3}m$), forming a three-dimensional (3D) structure (Fig. 2a). Li^+ ions occupy tetrahedral 8a sites, whereas Mn ions occupy octahedral 16d sites of the cubic close-packed arrangement of

the O^{2-} lattice in the spinel structure.^{19,45,46} Li^+ can migrate through vacant tetrahedral (8a) and octahedral sites (16c), which offer a three-dimensional diffusion network. The migration of Li^+ from and into the spinel cathode involves two notable voltage-plateaus. At a higher voltage (~ 4.0 V), Li^+ (de)intercalates from/into tetrahedral 8a sites, maintaining the cubic structure. On the other side, at a lower voltage (~ 3.0 V), Li^+ migrates from/into the vacant 16c octahedral sites. Close to this voltage range, $LiMn_2O_4$ undergoes an asymmetric lattice distortion resulting from the Jahn–Teller effect of high-spin $Mn^{3+}:3d^4(t_{2g}^3e_g^1)$.⁴⁷ This leads to the phase transition from a cubic structure to a tetragonal structure, which in turn results in rapid capacity fading. A two-phase region is often identified when $0.1 < x < 0.8$ in $Li_{x+1}Mn_2O_4$.⁴⁸ Hence, the use of this cathode is limited by a particular voltage range. Another major problem arises, when the electrolyte includes $LiPF_6$ salt with a trace amount of HF. This will trigger the Mn disproportionation reaction ($2Mn^{3+} \rightarrow Mn^{2+} + Mn^{4+}$), causing the dissolution of the cathode into the electrolyte on charge–discharge.⁴⁹

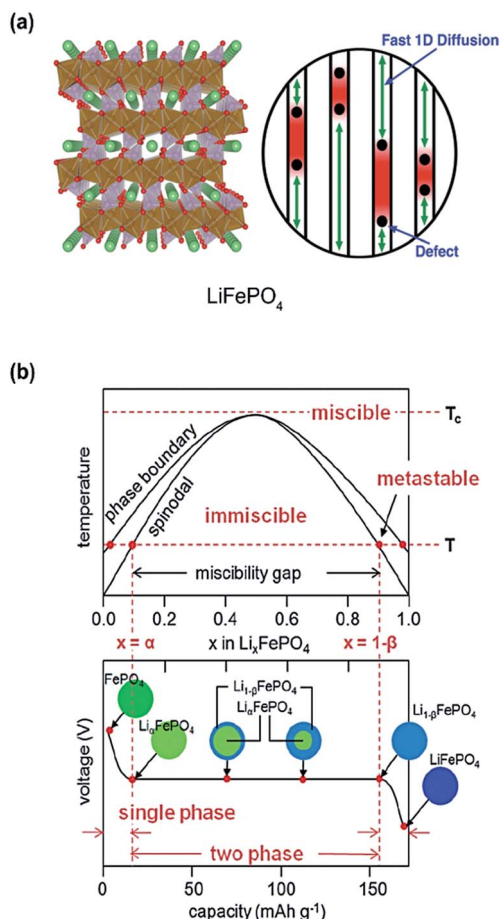


Fig. 4 (a) Crystal structure of $LiFePO_4$ with the diffusion pathway along the [010] direction. Illustration of point defects (Defect) that disturb Li^+ diffusion. Reproduced from ref. 206 with permission (Copyright 2010, American Chemical Society). (b) Two-phase region in the galvanostatic profile of $LiFePO_4$ and the corresponding spinodal-type phase diagram. Reproduced from ref. 207 with permission (Copyright 2011, Royal Society of Chemistry).

Another class of cathode is the olivine $LiFePO_4$ (LFP), which was introduced as a new type of cathode material belonging to phospho-olivines. Its reversible Li^+ extraction and insertion was firstly demonstrated in 1997 by Padhi and Goodenough *et al.*, revealing a redox potential of ~ 3.45 V vs. Li^0/Li^+ .⁵⁰ LFP is a polyanion material crystallized in an orthorhombic system (space group: $Pnma$) with a slightly distorted hexagonally close-packed (HCP) oxygen arrangement (Fig. 2a). P is located in tetrahedral sites, whereas Li^+ and Fe^{2+} occupy octahedral sites. The strong P–O bond in $(PO_4)^{3-}$ is the key factor that stabilizes the whole olivine structure, reducing O_2 release during charge and discharge. The major advantages of using the LFP cathode are the good thermal and electrochemical stability, environmental benignity, and cheaper material cost.⁵¹ However, this type of cathode suffers from low Li^+ diffusion and electronic conductivity, resulting in loss of capacity. The Li^+ diffusion channel of LFP is based on a one-dimensional (1D) channel, where the diffusion takes place along the edge-shared LiO_6 units. It is therefore that the Li^+ diffusivity in LFP can be easily affected by the point defects or impurities present in the 1D diffusion path (Fig. 4a). The electrochemical properties of LFP could be described by a flat plateau occurring at approximately 3.45 V in the galvanostatic profile (Fig. 4b). The plateau is characteristic of Li^+ (de)intercalation in the two-phase system of $LiFePO_4/FePO_4$, which is strongly associated with the cell performance.^{3,52,53} Thus, previous studies have attempted to explore the correct nature of the two-phase region of the LFP cathode to suggest better material designs or engineering treatments for the improvement of cell performance and stability. Various Li^+ (de)intercalation mechanisms that reasonably address the two-phase region were proposed, such as the shrinking-core model,⁵³ new-core shell model,⁵⁴ and domino-cascade model.⁵⁵

3. Advanced technologies

3.1 High voltage lithium-ion batteries

High voltage cathodes (4.5–5.0 V) have been considered as a potential candidate for advanced lithium-ion batteries. Spinel $LiMn_{1.5}Ni_{0.5}O_4$, olivine $LiCoPO_4$, olivine $LiNiPO_4$ and inverse spinel $LiNiVO_4$ are the major materials in this category.⁵⁶ They are designed to achieve higher charge capacity by enabling a deeper level of Li^+ deintercalated state. Among them, $LiMn_{1.5}Ni_{0.5}O_4$ demonstrates high cyclability with the voltage range of 4.7–4.8 V. It also reveals a fair thermal stability up to 250 °C.⁵⁷ One major problem of these high voltage cathodes is the severe decomposition of the electrolytes, as the conventional carbonate-based electrolytes become unstable in the voltage range higher than ~ 4.4 V.

3.2 High capacity lithium-ion batteries

In recent years, with the development of high-capacity lithium-ion cathodes, much attention has been devoted to the use of lithium-ion batteries in electric vehicles (EVs), and plug-in hybrid electric vehicles (HEVs/PHEVs). Particular attention is paid to the Li-rich (or excess) layered oxide cathode, often denoted as $xLi_2MnO_3 \cdot (1-x)LiTO_2$ ($T = Mn, Ni, Co$). A Li-rich

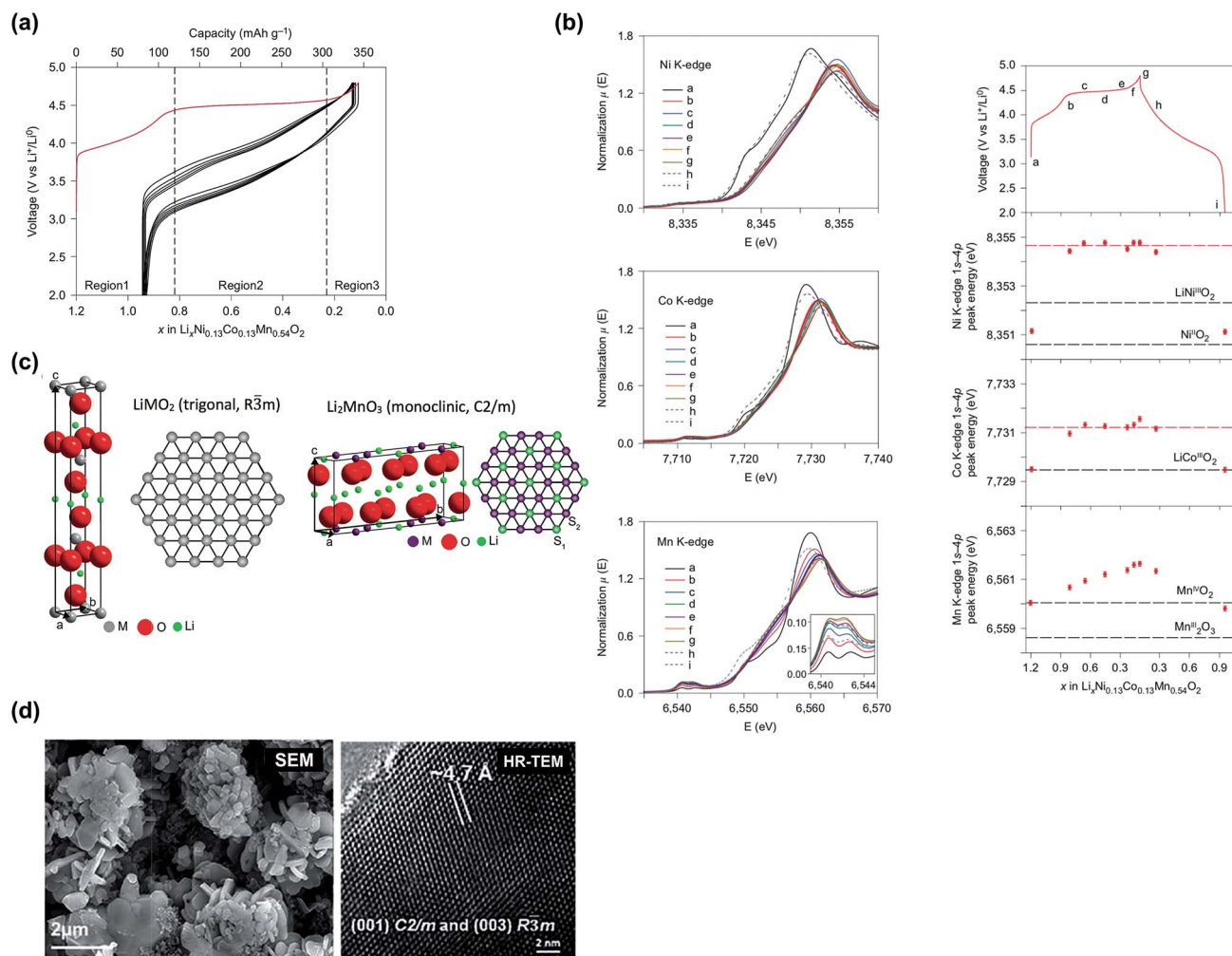


Fig. 5 (a) Galvanostatic profile of Li_{1.2}Ni_{0.13}Co_{0.13}Mn_{0.54}O₂ (Li-rich battery) with the cut-off voltage of 4.8 V (charge rate: 50 mA g⁻¹). The red plot indicates the first charge. Reproduced from ref. 208 with permission (Copyright 2016, Nature Publishing Group). (b) X-ray absorption near edge structure (XANES) spectra of the transition metal K-edge positions of Ni K-edge, Co K-edge, and Mn K-edge. The evolution of the corresponding oxidation states of Ni, Co, and Mn in the Li_{1.2}Ni_{0.13}Co_{0.13}Mn_{0.54}O₂ electrode. Reproduced from ref. 208 with permission (Copyright 2016, Nature Publishing Group). (c) LiMO₂ (trigonal, R $\bar{3}m$) where M = Ni, Co, and Mn, and Li₂MnO₃ (monoclinic, C2/m) unit cells and their atomic arrangements in the transition-metal layer. S₁ and S₂ represent site occupation by excess Li atoms and Mn atoms, respectively. Reproduced from ref. 209 with permission (Copyright 2016, American Chemical Society). (d) SEM image and high resolution transmission electron (HR-TEM) observation of the Li_{1.2}Co_{0.1}Mn_{0.55}Ni_{0.15}O₂ electrode, showing the distance (~4.7 Å) between two lattice fringes ((003) and (001)). Reproduced from ref. 63 with permission (Copyright 2013, Elsevier).

layered oxide cathode has been introduced by Thackeray *et al.*, demonstrating an integrated structure of the two distinct crystallographic phases between Li₂MnO₃ (monoclinic, C2/m) and LiTO₂ (trigonal, R $\bar{3}m$), where T represents the 3d-transition metals (Fig. 2b and 5c).^{58–60} The intimate integration is likely to be attributed to the interleaving of close-packed layers between the (001) planes from Li₂MnO₃ and the (003) planes from LiTO₂ with the interlayer spacing of ~4.7 Å (Fig. 5d).⁶⁰ The major advantages of the Li-rich layered transition metal oxide batteries are the superior theoretical capacity (>250 mA h g⁻¹) with high chemical potential (>4.6 V vs. Li⁰), high discharge capacity, good thermal stability, and lower cost compared to the traditional batteries (*e.g.* LCO, NMC, LFP).^{59–61} These high-energy density batteries, however, suffer from irreversible capacity loss (40–100 mA g⁻¹) during the initial cycle

(Fig. 5a).^{62–64} Voltage decay on further cycling is another major drawback of this battery system.^{65,66} The electrochemical properties of the Li-rich batteries can be characterized by a two-stage reaction in the galvanostatic profile during the initial electrochemical cycle. At a lower voltage, a slope can be identified up to ~4.5 V vs. Li/Li⁺, which is attributed to the redox reaction of Ni²⁺/Ni³⁺/Ni⁴⁺ and Co³⁺/Co⁴⁺ (Fig. 5b). At a higher voltage, a plateau like region occurs between 4.5 V and 4.8 V during the initial charge, where Li⁺ ions deintercalate from the Li₂MnO₃ component.^{60,67,68} The irreversible capacity loss during the initial cycle could be attributed to lithia (Li₂O) extraction from Li₂MnO₃, which in turn results in the formation of a layered MnO₂ component (Li₂MnO₃ → Li₂O + MnO₂).^{60,61,67} Similar to the NMC cathode, the Li-rich cathode undergoes a phase transition from a layered structure into a spinel structure on charge–

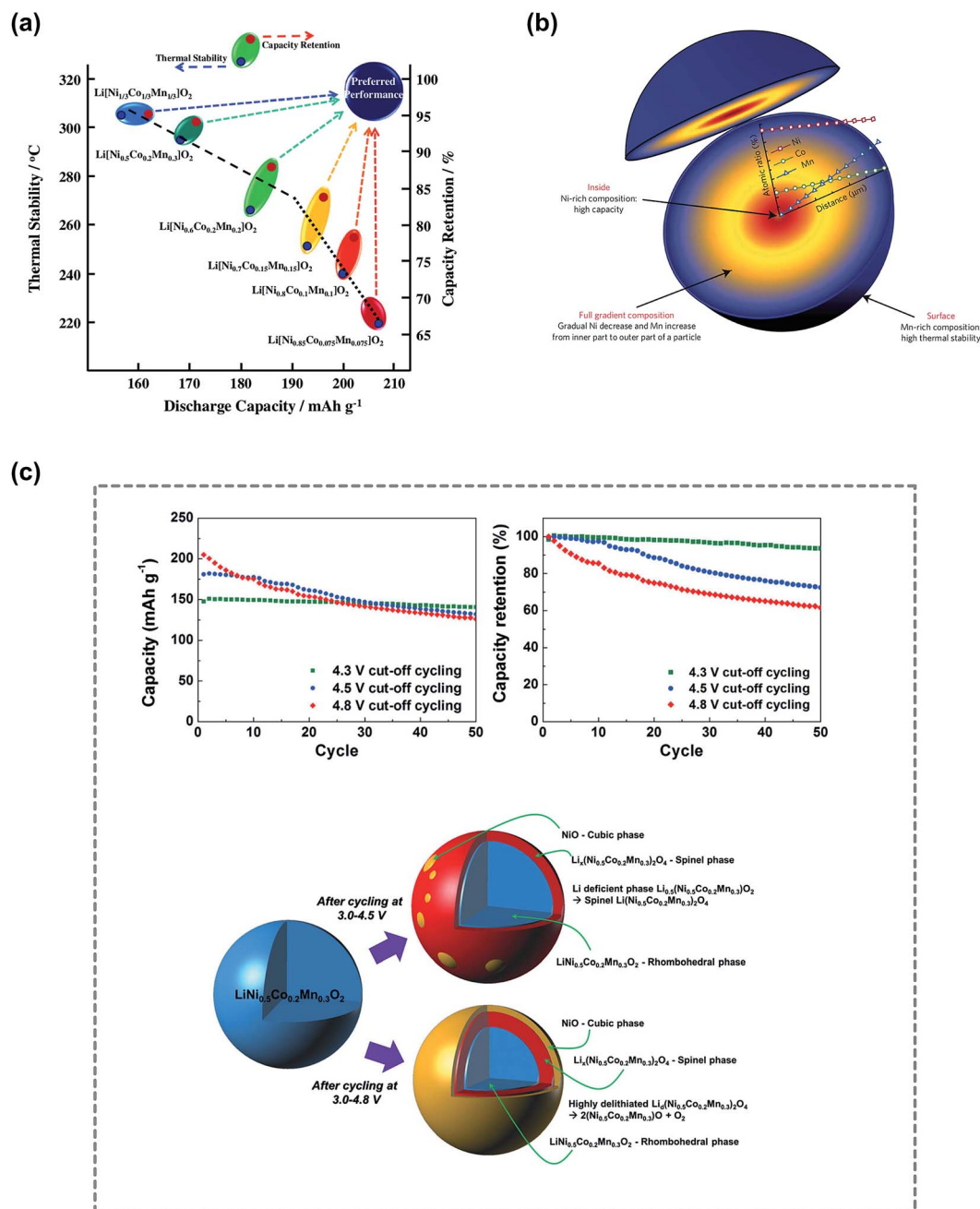


Fig. 6 (a) Plot of the relationship between thermal stability, discharge capacity and capacity retention for various types of Ni-rich cathodes. Reproduced from ref. 210 with permission (Copyright 2013, Elsevier). (b) Schematics of lithium transition-metal oxide particles with a Ni concentration gradient. Reproduced from ref. 72 with permission (Copyright 2012, Nature Publishing Group). (c) Cycling performance of LiNi_{0.5}Co_{0.2}Mn_{0.3}O₂ in the voltage range of 3.0–4.3 V, 3.0–4.5 V, and 3.0–4.8 V (charge rate: 0.4C), and its degradation mechanism. Reproduced from ref. 42 with permission (Copyright 2014, WILEY-VCH).

discharge.^{63,64,69,70} Especially, a previous study by Gu *et al.* reveals the origin of the phase transition.⁶⁹ It has been suggested that the spinel phase transition of LiMO₂ (M = Ni, Mn, Co, $R\bar{3}m$) is triggered by transition metal migration into the Li-layer. On the other hand, the phase transition to the spinel from Li₂MnO₃ ($C2/m$) is associated with lithia (Li₂O) removal, which in turn causes lattice strain in the electrode. Transition metal redox couples are the crucial factor that is strongly associated with the achievement of high charge capacities for lithium-ion

batteries. A recent study by Lee and Ceder *et al.* has reported a reversible Mn²⁺/Mn⁴⁺ double redox cathode of the Li-rich transition metal oxide battery.⁷¹ The introduction of high-valent cation (Nb⁵⁺, Ti⁴⁺) substitution with a partial replacement of O²⁻ by F results in a high charge capacity (>300 mA h g⁻¹) and energy density (~1000 W h kg⁻¹).⁷¹ The participation of Mn in the charge compensation mechanism not only enhances the battery capacity, but also offers advantages in terms of resource cost and environment issues.

Another class of the high capacity Li-ion battery is the Ni-rich layered oxide battery. Recent developments in the field of electric vehicles have led to a renewed interest in NMC cathodes with higher Ni concentration such as $\text{Li}[\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}]\text{O}_2$ (NMC532), $\text{Li}[\text{Ni}_{0.4}\text{Co}_{0.3}\text{Mn}_{0.3}]\text{O}_2$ (NMC433), gradient $\text{Li}[\text{Ni}_{0.86}\text{Co}_{0.1}\text{Mn}_{0.04}]\text{O}_2$, $\text{Li}[\text{Ni}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}]\text{O}_2$ (NMC622), $\text{LiNi}_{0.8}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$ (NCA) and $\text{Li}[\text{Ni}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}]\text{O}_2$ (NMC811), owing to their higher discharge capacities ($215\text{--}220\text{ mA h g}^{-1}$) and lower cost as compared to LCO and spinel batteries.^{56,72–74} The increase of the Ni concentration in the electrode composite significantly contributes to the increment of the charge capacities (Fig. 6a), due to the two-stage oxidation changes between $\text{Ni}^{2+}/\text{Ni}^{3+}$ and $\text{Ni}^{3+}/\text{Ni}^{4+}$. However, the high voltage ($\sim 4.5\text{--}4.8\text{ V}$) required for the charge compensation mechanism of the Ni-rich cathode leads to a severe degradation process.^{42,43,75} The aggressive capacity fading is typically connected to the NiO surface reconstruction or clustering on the Ni-rich active material particles (Fig. 6c).⁴² This involves not only rapid electrolyte decomposition, but also phase transition from a layered ($R\bar{3}m$) phase into a spinel-like and/or rock-salt phase on charge and discharge.^{42,75} To counter the structural mismatch, Sun *et al.* demonstrated a nanostructured high-energy cathode with a concentration gradient between Ni, Mn, and Co (Fig. 6b).⁷²

4. State-of-the-art technologies

4.1 Li–sulfur batteries

As a next generation battery technology, sulfur (S) is a promising candidate for cathode materials (Fig. 7c). It offers a very high theoretical capacity of 1675 mA h g^{-1} , about 10 times that of current transition metal oxides. It has higher abundance than cobalt and nickel in the earth's crust, and is of very low cost.^{76,77} The schematic of the working principle of a Li–S battery and its galvanostatic profile are shown in Fig. 7a and b. In spite of these significant advantages, a number of technical challenges hinder the further applications of sulfur cathodes. They suffer from low electrochemical potential vs. Li/Li^+ , low electrical conductivity, and like other conversion type electrodes, sulfur suffers from a significant volume expansion of $\sim 80\%$, causing material cracking and detachment from the electrode conductive network.^{78,79} Also, the sulfur based cathode is of low vaporization temperature, leading to a pure sulfur loss during the electrode drying process under vacuum.⁸⁰ In addition, intermediate reaction products formed during charge and discharge, *i.e.*, polysulfide Li_2S_n ($3 \leq n \leq 8$), are soluble in the electrolyte, and thus lead to the shuttle mechanism and capacity loss.^{81–84}

The long-chain polysulfides dissolved in the electrolyte will migrate back to the anode, and they will be reduced into lower order polysulfides. This process is followed by the migration of the lower order polysulfides back to the cathode, resulting in a re-oxidation. The shuttle effect will produce insoluble products such as Li_2S_2 and Li_2S , causing self-discharge and overall poor coulombic efficiency. These technical barriers prevent the mass commercialization of sulfur cathodes for Li–S batteries despite decades of research.

A number of studies have attempted to address the technical barriers of Li–S batteries. Luo *et al.* reported oxygen-stabilized

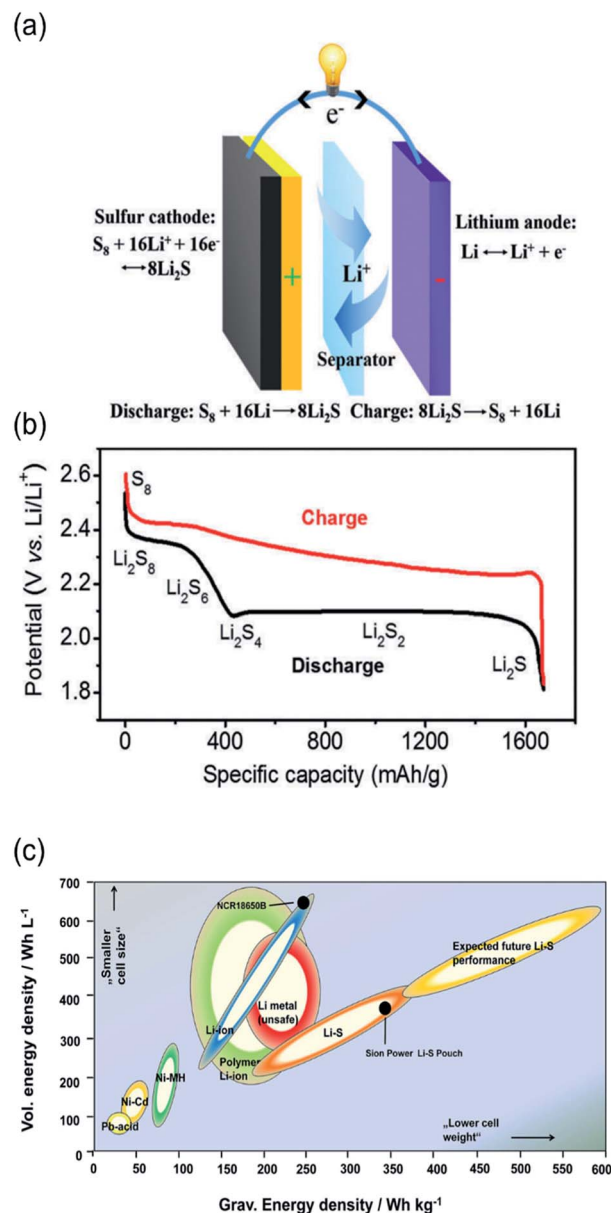


Fig. 7 (a) Illustration of the Li–S battery working principle with redox reactions. Reproduced from ref. 202 with permission (Copyright 2015, Royal Society of Chemistry). (b) Galvanostatic profile of the Li–S battery with multiple reduction stages. Reproduced from ref. 202 with permission (Copyright 2015, Royal Society of Chemistry). (c) Comparison of energy densities between the Li–S battery and various energy storage systems. Reproduced from ref. 211 with permission (Copyright 2015, WILEY-VCH).

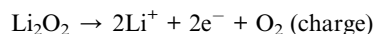
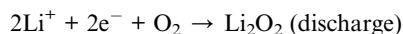
sulfur in a carbon matrix, showing good reversible capacity with high cycle numbers for Li–S and Na–S batteries.⁸⁵ The performance improvement of the Li–S battery can be explained by the suppression of the polysulfide intermediates owing to the entrapped sulfur in the carbon matrix. Another study has shown the development of a carbonized mesoporous wood fiber as a host material to intercalate sulfur, and thereby providing high cycling stability for Li–S batteries.⁸⁶ A unique approach to suppress the shuttle effect of LiS batteries has also been shown,

using selenium (Se) doping with the formation of a stable solid-electrolyte interphase (SEI).⁸⁷

4.2 Li-air batteries

Lithium-air battery (Li-O₂) with an organic electrolyte was demonstrated by Abraham *et al.* in 1996 (Li-O₂).⁸⁸ The working principle of the Li-O₂ battery can be explained by the redox reactions between Li metal and oxygen. Owing to their superior theoretical energy density ($\sim 3500 \text{ W h kg}^{-1}$), lithium-air batteries have drawn much attention, as an alternative technology to high-capacity lithium-ion batteries.^{76,89–91} Their practical energy density can reach 1700 W h kg^{-1} .⁹⁰ The anode of a Li-air battery is typically Li metal, whereas the cathode is a carbon matrix with a catalyst. This battery can be categorized into four different architectures on the basis of its electrolyte type, such as Li-O₂ aprotic, Li-O₂ aqueous, Li-O₂ mixed and Li-O₂ all-solid-state batteries.⁹⁰ Among them, much attention is paid to the aprotic configuration, due to the instant formation of a barrier between the electrolyte and anode, which in turn protects the Li-metal from the redox reaction with the electrolyte on charge and discharge.

Fig. 8 shows a schematic cell configuration for the Li-O₂ batteries with the following reactions:



During the discharging process, oxygen is reduced to Li₂O₂, providing an electromotive force of 2.9 V with a theoretical specific energy density of 3500 W h kg^{-1} ,^{92,93} when the cell is configured with a Li-metal anode. Also, the simplicity in the active material composite makes Li-O₂ batteries available with lower cost as compared with the conventional lithium-ion batteries. The Li-air battery can be operated both with aqueous and non-aqueous electrolytes. Under aqueous conditions, during the discharge, the formation of LiOH can prevent the air cathode from clogging and protect it from being affected by over-potentials. On the other hand, the LiOH cannot be fully decomposed on charge and thereby it will precipitate, leading to lower energy density and poor cyclability.^{76,92}

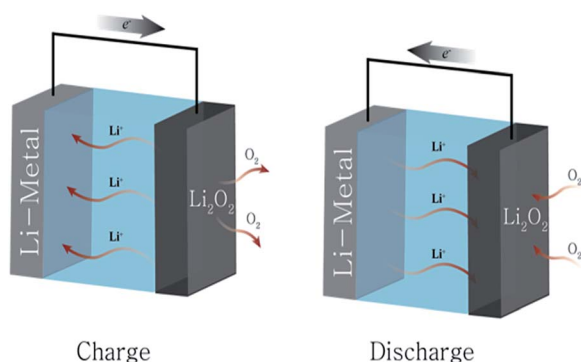


Fig. 8 Schematics of cell configuration and working process of the Li-O₂ battery.

The aqueous cells require a surface protecting or lithium ion conducting layer such as LISICON-type glass ceramic (lithium superionic conductor, $\text{Li}_{(1+x+y)}\text{Al}_x\text{Ti}_{2-x}\text{Si}_y\text{P}_{(3-y)}\text{O}_{12}$) to prevent direct contact of the Li metal surface with H₂O.⁷⁶ These treatments with surface coating layers, however, do not sustain long enough to hinder the formation of lithium dendrites.⁹² To tackle this issue, it is necessary not only to improve the decomposition rate of lithium peroxide (Li₂O₂), but also minimize the side reactions from the Li metal. Also, the use of ambient air in the presence of N₂, CO₂, and H₂O vapor can lead to a series of side reactions, causing poor reversibility of the electrochemical performance.⁹⁴ Research based on organic electrolytes has been investigated for several years with regard to the present Li-ion battery technology, but the electrolytes from Li-ion batteries cannot be directly utilized in Li-O₂ batteries, because of the stability issue with the electrode. Aprotic electrolytes for the Li-O₂ batteries should be stable against the active redox reactions that lead to the formation of reduced oxygen species. It also requires high O₂ solubility and diffusivity to enhance mass transport rates to the cathode, as well as low volatility and high boiling point to minimize the solvent evaporation at the cathode.⁹⁵ In the beginning, carbonate based liquid electrolyte solvents, such as propylene carbonate (PC), ethylene carbonate (EC), and dimethyl carbonate (DMC), were widely used for lithium-air batteries, but the cycling performance was significantly poor.^{89,91,96–98}

Further investigations have revealed that the carbonate-based electrolytes are susceptible to nucleophilic attack, and they can be easily decomposed in the presence of O₂ reduced species on discharge. As a consequence, during discharge, lithium carbonate (Li₂CO₃) and alkyl carbonate (RO-(C=O)-OLi) are formed instead of lithium peroxide (Li₂O₂).^{98–101} Alternatively, sulfur-containing solvents such as dimethylsulfoxide (DMSO), sulfolane and ether-based electrolytes have also been broadly employed for Li-O₂ batteries, especially, DMSO possesses a high oxidation potential (4.8 V vs. Li⁺/Li) with a high boiling point (189 °C) and weak acidic properties.^{89,102,103} The performance of Li-air batteries with DMSO and ether-based electrolytes is believed to be highly reversible. However, the DMSO based electrolytes are not only electrochemically unstable when in contact with lithium metal, but also generate side products, such as Li₂CO₃, HCO₂Li and CH₃CO₂Li. Recent studies have demonstrated that non-aqueous solvents with lithium-ions are reactive toward the reduced O₂ products.^{104–106} A previous study by Qiao *et al.* introduced a DMSO-solvated contact ion pair (CIP) electrolyte system composed of [Li(DMSO)₃]⁺[TFSI][−].¹⁰⁷ This group has shown a Li-O₂/CO₂ battery, which can be operated at a very low charge potential (3.5 V) by stabilizing peroxodicarbonate (C₂O₆^{2−}). In the presence of CO₂, peroxodicarbonate (C₂O₆^{2−}) forms during the discharge process, and readily evolves to Li₂CO₃ with the existence of lithium cations. With the increment of the salt concentration, they identified the presence of [Li(DMSO)₃]⁺[TFSI][−] by Raman spectroscopy (Fig. 9). The formation of C₂O₆^{2−}, which could be stabilized during cycling, contributes to the good rate capability and cyclability of the Li-O₂/CO₂ battery system. It has also been suggested that the decomposed TFSI[−] anion stabilizes the solid electrolyte

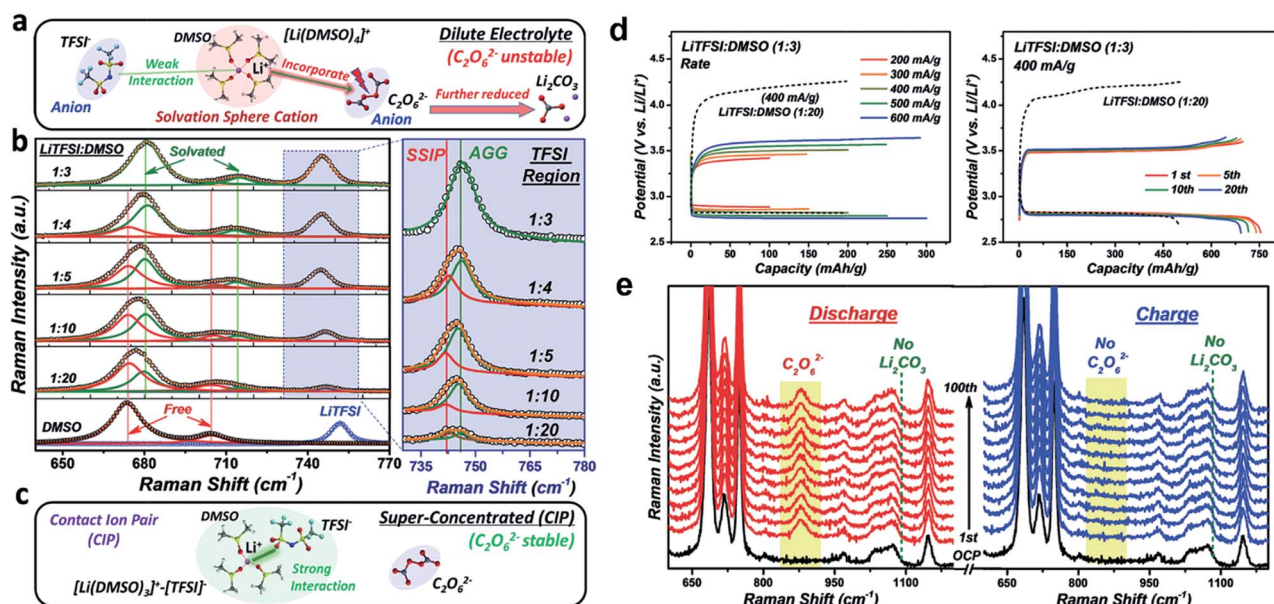


Fig. 9 Design of the electrolyte structure. (a) Schematic illustrations of the electrolyte structure in a dilute LiTFSI/DMSO-based Li-O₂/CO₂ cell, and relevant discharged components. As an intermediate, peroxodicarbonate (C₂O₆²⁻) coordinates with solvated Li ions and dissociates into Li₂CO₃ as the typical final product. (b) Raman spectra of LiTFSI/DMSO solutions with various mole ratios. Spectra are fitted by Gaussian-Lorentzian distribution functions, and the SNS stretch region in TFSI⁻ (blue outlined) is magnified for clarity. In the LiTFSI/DMSO electrolyte with a 1 : 3 mole ratio, neither free DMSO molecules nor solvent-separated ion pairs (SSIPs) were observed. Instead, sheathed by DMSO molecules, Li⁺ and TFSI⁻ were completely aggregated (AGG) within a contact ion pair (CIP) fluid network. (c) Schematic illustration of the newly introduced super-concentrated fluid network electrolyte composed of [Li(DMSO)₃]⁺–[TFSI]⁻. Reproduced with permission.²⁷ Electrochemical performance of Li-O₂/CO₂ batteries. Galvanostatic voltage profiles of Li-O₂/CO₂ (1 : 1, v/v) cells with a specific LiTFSI/DMSO (1 : 3, mole ratio) electrolyte. (d) Initial cycle at current densities from 200 to 600 mA g⁻¹ (left). Full discharge-charge profiles over 20 cycles controlled by potential limits from 2.6 to 3.65 V at 400 mA g⁻¹ (right). (e) In situ Raman spectra collected at the end of each discharge state (red) and charge (blue) state among typical cycles during cycling with a fixed capacity of 200 mA h g⁻¹ (at 400 mA g⁻¹ current density). Reproduced from ref. 107 with permission (Copyright 2018, Royal Society of Chemistry).

interphase by mitigating the attack from superoxide. This will lower the charging overpotential of the Li-O₂ battery.^{108,109}

There are also key issues that need to be addressed at the oxygen electrode. The oxygen electrode in Li-air battery requires high electronic and ionic conductivity, electrochemical stability, fast oxygen diffusivity, porous structure to store the discharged product Li₂O₂, and high activity for the OER (oxygen evolution reaction) and ORR (oxygen reduction reaction) processes. There are a wide range of carbon materials utilized as electrodes in Li-air batteries, such as carbon black, carbon fiber, carbon nanotubes, rGO and graphene-based materials.^{110–112} Unfortunately, all carbonaceous electrode materials can notably degrade by the interactions with superoxide and peroxide moieties. Most recently, therefore, non-carbonic oxygen electrodes have been of great interest. Non-carbonaceous materials such as Au,⁸⁹ nano-porous Au,⁸⁹ TiC,¹¹³ Ti₄O₇,¹¹⁴ Ru,⁹² RuO₂,¹¹⁵ MoS₂,¹¹⁶ Mo₂C (ref. 117) and Co₃O₄ (ref. 118) are widely studied for the oxygen electrode. These metals and metal oxides exhibit excellent stability with little oxygen reduction. Asadi *et al.* showed that MoS₂ nano-flakes with the ionic liquid (IL) 1-ethyl-3-methylimidazolium tetrafluoroborate (EMIM-BF₄) can serve as an effective cocatalyst for discharge and charge processes in Li-O₂ batteries.¹¹⁶ The MoS₂/IL system shows superior reaction rates as well as lower overpotentials for the oxygen reduction and evolution reactions,

as compared to Au and Pt metals. The improvement mechanism by MoS₂/IL has been revealed with the DFT calculations. EMIM⁺ tends to form isolated Mo sites which prevent O₂ dissociation and enable oxygen reduction. The ionic liquid EMIM-BF₄ facilitates the formation of Li₂O₂ by the dissolution of superoxide species (O₂⁻). The performance of the MoS₂/IL cathode/electrolyte (0.1 M LiTFSI) system is tested by a capacity-limited (500 mA h g⁻¹) cycling up to 50 cycles at a current density of 0.1 mA cm⁻². The discharge begins at 2.9 V and the target capacitance is obtained at a potential 2.69 V. The charging process is completed at a potential of 3.49 V, showing a discharge/charge polarization gap of 0.8 V. This increased to 0.85 V after 10 cycles, and the further discharge capacity of MoS₂/IL was ~1250 mA h g⁻¹. To develop Li-O₂ batteries with long-term stability, the electrochemical stability of DMSO should be improved. Moreover, additional surface passivation layers would be required to prevent the Li metal from degrading *via* side reactions.

4.3 Organic electrode batteries

Using organic materials as electrodes for lithium-ion batteries has been proposed over the last few decades, but they have drawn less attention than the metal based electrodes.^{119,120} This battery type benefits from good sustainability, eco-friendly feature, flexibility in material design, and resource abundance, owing to the

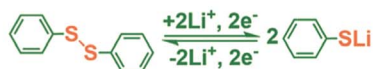
presence of lightweight elements such as C, H, N and O.^{119,120} Several types of organic materials, *i.e.*, organosulfurs, carbonyl compounds, organic radicals (or radical polymers), and imine groups (C=N) have been vastly studied in the past two decades.^{119,121} In particular, carbonyl based compounds (*e.g.* quinone) are of great interest because of their high redox activity, high redox potential, and abundance. Their redox activity with reversible electron transfer could be extended to multi-electron reactions when additional carbonyl groups are conjugated, forming multivalent anions.¹²⁰ In 1972, Alt *et al.* reported the utilization of quinone compounds as a cathode for secondary batteries.¹²² Chloranil was used as an active material for the battery, showing 95% capacity retention after 50 electrochemical cycles (current rate: 0.5C).¹²² However, batteries with organic electrode materials also have major drawbacks such as low-energy density, poor conductivity, poor reaction kinetics, and high solubility in electrolytes.^{119,120,123} Common strategies to overcome these problems are addition of conductive additives (*e.g.* carbon), modification of the structure with conductive materials (*e.g.* carbon nanotubes or graphene), and tuning the electrode-level.^{119,123} Previous studies by Luo *et al.* have suggested azo (N=N) functional groups as alternative organic materials to tackle the aforementioned drawbacks (Fig. 10).^{121,124} Whereas the carbonyl groups or imine groups require more than two functional groups, the azo based electrode only needs one functional group, which in turn simplifies the chemical structure.¹²⁴ The batteries with azo compounds exhibit good cycling stability, as well as high rate capability.¹²¹ Employing self-assembled organic nanowires using croconic acid disodium salt (CADs) is another notable approach that can deal with the inherent problems of the organic electrodes for lithium-ion batteries.¹²⁵

4.4 Solid-state batteries

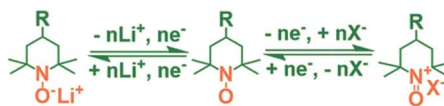
Solid electrolytes are the key component that enables the development of solid-state batteries. They can be divided into three main categories such as inorganic solid-electrolytes, solid polymer electrolytes, and thin-film solid-electrolytes.¹²⁶ Garnets (*e.g.* $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$),¹²⁷ NASICON-type (*e.g.* $\text{LiZr}_2(\text{PO}_4)_3$ (LZP), $\text{LiTi}_2(\text{PO}_4)_3$ (LTP), and $\text{LiGe}_2(\text{PO}_4)_3$ (LGP)),^{128,129} sulfide-type ($\text{Li}_2\text{S-SiS}_2$ -based or $\text{Li}_2\text{S-P}_2\text{S}_5$ -based)^{130,131} and perovskite-structured ($\text{Li}_{3-x}\text{La}_{2/3-x}\text{TiO}_3$)^{132,133} are the widely studied inorganic solid electrolytes. Among them, oxide-type materials (garnets, NASICON, and perovskite) have been of great interest as they offer certain merits for the all-solid-state batteries, *i.e.*, high chemical stability, excellent mechanical properties, and good safety with reasonable conductivities (10^{-4} to $10^{-3} \text{ S cm}^{-1}$).¹²⁶ In spite of the advantages, their non-flexibility and high-cost for scale-up productions hindered their successful commercialization. On the other hand, sulfide-type solid-electrolytes (Fig. 11) have high conductivity ($10^{-2} \text{ S cm}^{-1}$), good mechanical properties and flexibility, and low grain boundary resistance.^{126,134} However, the drawbacks of this material are the low oxidation stability and poor compatibility with electrodes.^{126,134} In addition, the instability when exposed to moisture, which in turn generates H_2S gas, also limits the lifespan of this battery application.¹³⁴ Solid polymer electrolytes (SPEs) have also been considered as promising solid-electrolytes for solid-state batteries. Polymer electrolytes commonly employed polyethylene oxides (PEOs),^{135,136} poly(acrylonitrile) (PAN),¹³⁷ polyvinylidene fluoride (PVDF),¹³⁸ and poly(methyl methacrylate) (PMMA)¹³⁹ for the host material. In particular, PEO with the repeat unit of $-\text{CH}_2-\text{CH}_2-\text{O}-$ containing lithium salts has received considerable attention due to its non-toxic

Conventional Reaction

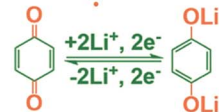
Type 1 S-S Reaction



Type 2 Doping Reaction



Type 3 C=O Reaction



Type 4 C=N Reaction



New Reaction

Type 5 N=N Reaction

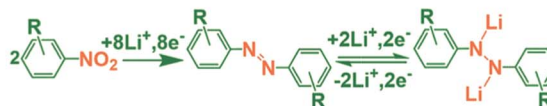


Fig. 10 Comparison of organic molecules for organic electrode lithium-ion batteries. The N=N reaction indicates the reduction of two nitro groups by lithium-ions to yield an azo group with the formation of Li_2O . Reproduced from ref. 121 with permission (Copyright 2018, WILEY-VCH).

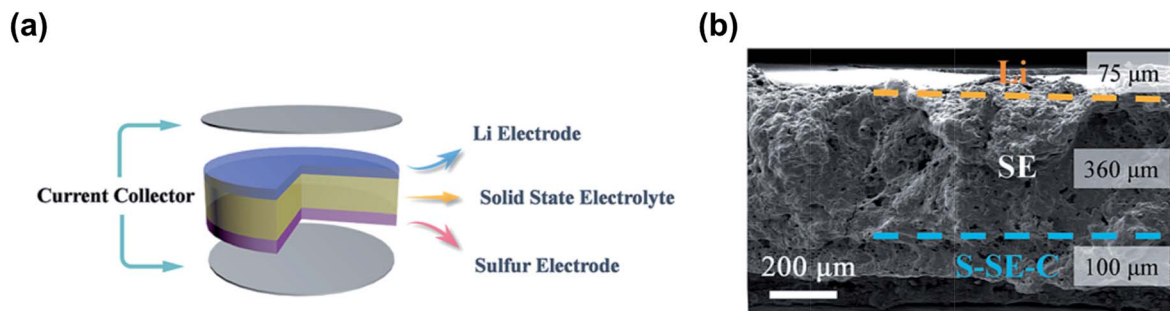
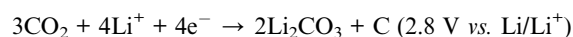


Fig. 11 (a) Illustration of an all-solid-state Li-sulfur battery. (b) SEM observation of the cross-section of an all-solid-state Li-sulfur battery with the $\text{Li}_7\text{P}_{2.9}\text{-S}_{10.85}\text{Mo}_{0.01}$ electrolyte (SE). Reproduced from ref. 212 with permission (Copyright 2017, Royal Society of Chemistry).

property, low-cost, and good chemical stability.¹³⁶ The bonding of Li^+ in PEO with chain segment dynamics was identified in previous studies.^{140,141} Li^+ can migrate along the polymer chains and/or motion with PEO segments.¹⁴¹ However, the lithium-ion conductivity ($\sim 10^{-7} \text{ S cm}^{-1}$) at room temperature is not only lower than those of the carbonate electrolytes, but also those of inorganic solid-electrolytes.¹⁴² Improvement of conductivity in the solid-electrolytes was therefore a key issue for a long time, and it still remains unclear whether crystalline or amorphous structures can enhance the ion conductivity. Using polymers as solid-electrolytes has the common advantages of good flexibility, high stability with Li-metal, and low shear modulus, but they suffer from low thermal stability, low ionic conductivity, and low oxidation chemical potential. These issues should be challenged for the practical use of the solid polymer electrolytes (SPEs). Thin-film solid electrolytes have also drawn attention as they lead to the development of thin-film batteries that have higher energy density, higher flexibility in application design, and higher compactness as compared to the conventional lithium-ion batteries. The thin-film solid electrolyte for secondary lithium batteries was first demonstrated in the early 1980's using the $\text{Li/Li}_{3.6}\text{Si}_{0.6}\text{P}_{0.4}\text{O}_4/\text{TiS}_2$ composite.¹⁴³ There are various methods that can be used for thin-film deposition such as radio frequency sputtering (RFS),¹⁴⁴ pulsed-laser deposition (PLD),¹⁴⁵ chemical vapor deposition (CVD),¹⁴⁶ and atomic layer deposition (ALD).^{147,148} One of the extensively examined thin-film batteries is the lithium phosphonitroxide glass (LiPON) battery. This application was initially reported by Bates *et al.* in 1992,¹⁴⁹ and further studied by a group in Oak Ridge National Laboratory (ORNL, U.S.).¹⁵⁰ LiPON could be fabricated by RF-magnetron sputtering using Li_3PO_4 as a target in nitrogen gas. The main advantage of using LiPON is the good stability with lithium metal and a number of cathode materials. It has been reported that the stability window of LiPON was between 0 V and 5.5 V with respect to the Li-metal electrode.¹⁵¹ LiPON also benefits from its good hardness and thermal stability.¹⁵² Hence, the thin-film LiPON battery ($\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4/\text{LiPON}/\text{Li}$) could provide a cycle life of 10 000 cycles with the capacity retention of 90% up to 5 V.¹⁵³ LiPON has recently become a commonly used solid-electrolyte for the all-solid-state thin-film batteries, but its relatively lower conductivity than oxide-based materials must be overcome to develop high-capacity thin-film batteries.

4.5 Li-CO₂ batteries

In recent years, the battery community has shown an increased interest in Li-CO₂ batteries. This battery technology enables carbon dioxide (CO₂) fixation while storing energy. Previous research by Takechi *et al.* (2011) revealed a novel battery employing gases, namely, $\text{Li}/\text{CO}_2\text{-O}_2$ exhibited a high irreversible discharge capacity, which is three times as much as the capacity of a Li-air battery.¹⁵⁴ Later on, Lim and Kang *et al.* developed a reversible $\text{Li}/\text{CO}_2\text{-O}_2$ battery using dimethyl sulfoxide (DMSO) that leads to the reversible formation of Li_2CO_3 .¹⁵⁵ Whereas the previous Li gas batteries utilized a CO₂ and O₂ gas mixture, Xu and Archer *et al.* have introduced a primary Li-CO₂ battery that employs pure CO₂ gas as the working electrode for achieving charge capacities.¹⁵⁶ A year later, Liu *et al.* demonstrated a pure and reversible Li/CO_2 battery using LiCF_3SO_3 in TEGDME (1 : 4 in mole) as the electrolyte.¹⁵⁷ The charge capacity of the Li-CO₂ can be achieved by the following electrochemical reaction:^{157,158}



Li_2CO_3 is the key component that determines the reversibility of this battery application. However, Li_2CO_3 is an insulator and thus requires a high over-potential to be oxidized on charge. Recent studies by Zhou's group have utilized ruthenium (Ru) as a cathode catalyst to effectively reduce the high over-potential, thereby improving the reversibility of the Li-CO₂ battery (Fig. 12).^{158,159} Li/CO_2 battery is undoubtedly an attractive technology as it can simultaneously deal with energy and environmental issues. However, high over-potential and low coulombic efficiency are the major problems that need to be tackled for practical use.

5. Towards hybridization with Li-ion batteries

5.1 Application of metal halide perovskite

Metal halide perovskite (MHP) refers to a class of semi-conducting materials that can be represented by the general formula of ABX_3 . MHPs adopt a similar crystal structure to oxide perovskites such as calcium titanate (CaTiO_3). The halide

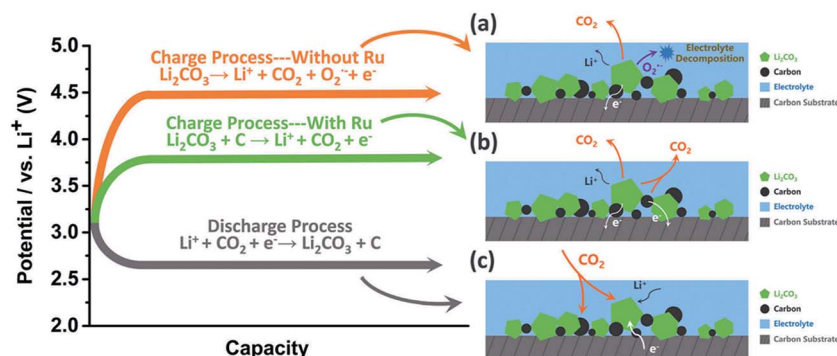


Fig. 12 Schematics of the (a) reaction mechanism of a Li–CO₂ battery on charge without a Ru catalyst, (b) reaction mechanism of the Li–CO₂ battery on charge with a Ru catalyst, and (c) discharging process. Reproduced from ref. 159 with permission (Copyright 2017, Royal Society of Chemistry).

anions (X = I[−], Br[−], Cl[−]) and metal cations (B = Pb²⁺, Sn²⁺) form the BX₆ octahedron (Fig. 10). The corner-shared BX₆ octahedra extend to a three-dimensional (3D) network stabilized by the presence of larger cations (A being larger than B; A⁺ = CH₃NH₃⁺, NH₂CH=NH₂⁺, Cs⁺, etc.). MHPs received tremendous attention in the past years after their first successful application in photovoltaic (PV) devices in 2009.^{160–163} In 2017, Tsutomu Miyasaka (Toin University of Yokohama, Japan), Nam-Gyu Park (Sungkyunkwan University, Republic of Korea), and Henry Snaith (University of Oxford, United Kingdom) received recognition as the 2017 Citation Laureates by Clarivate Analytics for their discovery and application of perovskite materials to achieve efficient energy conversion.¹⁶⁴ Since the successful demonstration in PV technology, applications of MHPs have been rapidly extended to other technologies, such as light-emitting diodes (LEDs),¹⁶⁵ lasers,¹⁶⁶ X-ray detectors and imaging for medical applications,^{167,168} field effect transistors (FETs),^{169,170} etc. A few recent studies suggest that MHPs also have potential for battery applications (Table 1).^{171–180}

For example, the room temperature chemical diffusion coefficient of lithium ions (Li⁺) within the CH₃NH₃PbBr₃ perovskite lattice exhibits a value as high as $D_{\mu} \sim 10^{-7} \text{ cm}^2 \text{ s}^{-1}$ meaning that the ionic conductivity is in the order of $10^{-3} \Omega^{-1} \text{ cm}^{-1}$ for highly lithiated electrodes.¹⁸¹ As a comparison, graphitic carbon and LiC₆ (resulting from Li⁺ interaction with graphite employed as the anode in conventional Li ion batteries) have $D_{\mu} \sim 10^{-12}$ – $10^{-5} \text{ cm}^2 \text{ s}^{-1}$ and $D_{\mu} \sim 10^{-8}$ – $10^{-10} \text{ cm}^2 \text{ s}^{-1}$, respectively.^{182,183} Xia *et al.*¹⁷⁴ applied for the first time CH₃NH₃PbBr₃ and CH₃NH₃PbI₃ organic–inorganic hybrid perovskites in Li-ion batteries as the anode material. The first discharge capacity for CH₃NH₃PbBr₃ was 331.8 mA h g^{−1} at a current density of 200 mA g^{−1} (Table 1). This capacity is approximately the same compared to that of commercially available graphite. The capacity of CH₃NH₃PbBr₃ decreased rapidly in the first 30 cycles, but after that it decreased slowly from 157.4 mA h g^{−1} (31st cycle) to 121 mA h g^{−1} (200th cycle) with relative retention of 76.9% in the subsequent 170 cycles. Vincente and Garcia-Belmonte¹⁷⁶ also employed CH₃NH₃PbBr₃ and reported that long-term specific capacity attains values

approaching 200 mA h g^{−1} (Table 1). The microscopic views of how Li ions interact with a hybrid perovskite structure during the charging–discharging process are still under intense debate.^{175,176,178} The fundamental microscopic understanding will play a major role in providing insights to questions such as (i) why CH₃NH₃PbI₃ perovskites show lower discharge capacity compared to that of CH₃NH₃PbBr₃, *i.e.*, what role do halides play in the Li-ion storage mechanism? (ii) What are the physico-chemical dynamical processes involved during charging–discharging in MHP materials? A maximum theoretical capacity of 55.96 mA h g^{−1} was calculated for CH₃NH₃PbBr₃ assuming that one Li⁺ could intercalate per formula unit.¹⁷⁸ However, the above reported experimentally measured discharge capacities are approximately four¹⁷⁶ to six¹⁷⁴ times larger than the theoretical capacity. It has been proposed that reversible Li⁺ storage takes place by Li ion intercalation into the host electrode without severe distortion and degradation of the CH₃NH₃PbBr₃ lattice.¹⁷⁶ Lithiation reaching molar contents of $x = 3$ in Li_xCH₃NH₃PbBr₃ was reported (Fig. 13).^{176,178} Alternatively, Dawson *et al.*¹⁷⁸ proposed that conversion reactions with Li are more energetically favorable in CH₃NH₃PbX₃ (X = I, Br, Cl) than Li intercalation. The authors carried out density functional theory (DFT) calculations to simulate four primary Li concentrations, $x = 0.037, 0.125, 0.5$, and 1.0 in Li_xCH₃NH₃PbX₃, where $x = 1.0$ represents full Li intercalation. At a low Li concentration, insertion of Li-ions into CH₃NH₃PbI₃ is more favorable than that for other two perovskites (CH₃NH₃PbBr₃ and CH₃NH₃PbCl₃). At full Li intercalation ($x = 1.0$), the PbX₆ octahedra undergo structural distortion in all three MHPs. In particular, the distortion strength is more pronounced in CH₃NH₃PbBr₃ and CH₃NH₃PbCl₃, which suggest the existence of conversion or decomposition reactions when approaching full Li intercalation. On the basis of electrochemical characterization, X-ray diffraction (XRD), and DFT, two potential conversion reaction mechanisms were proposed:¹⁷⁸

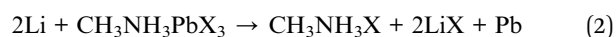
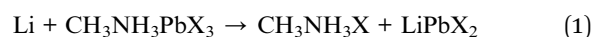


Table 1 Summary of metal halide perovskite-based Li-ion batteries. The electrochemical performances were tested in a standard CR2032 coin cell configuration if there is no statement. Abbreviations: PVDF = poly(vinylidene fluoride); EC = ethylene carbonate; EMC = ethylmethyl carbonate; DMC = dimethyl carbonate; DEC = diethyl carbonate; FEC = fluoroethylene carbonate; MA = CH₃NH₃; PA = CH₃(CH₂)₂NH₃; AB = acetylene black; EDBe = 2,2'-(ethylenedioxy)bis(ethylammonium); Li(TFSI) = Li bis(trifluoromethane)sulfonimide

Structure				
Anode	Electrolyte	Cathode	Discharge capacity (mA h g ⁻¹)	Ref.
CH ₃ NH ₃ PbI ₃ ; Conductive carbon black: PVDF binder (80 : 10 : 10 mass ratio) on copper foil	1 M LiPF ₆ in EC + EMC + DMC (1 : 1:1 vol ratio)	Li-foil	43.6 (1 st cycle; 200 mA g ⁻¹)	ref. 174
CH ₃ NH ₃ PbBr ₃ ; Conductive carbon black: PVDF binder (80 : 10 : 10 mass ratio) on copper foil			331.8 (1 st cycle; 200 mA g ⁻¹)	
CH ₃ NH ₃ PbBr ₃ ; Conductive carbon black: PVDF binder (80 : 10 : 10 mass ratio) on copper foil	1 M LiPF ₆ in EC + EMC + DMC (1 : 1:1 vol ratio)	Li-foil	~550 (1 st cycle; 200 mA g ⁻¹); 150–200 (after 10 cycles)	ref. 176
CH ₃ NH ₃ PbBr ₃ ; Carbon super P Li: PVDF binder (80 : 10 : 10 mass ratio) on copper foil	1 M LiPF ₆ in EC + DMC (1 : 1 wt)	Li-foil	134.3 (20 mA g ⁻¹)	ref. 178
CH ₃ NH ₃ PbBr ₃ /ITO	Three-electrode cell (0.1 mA cm ⁻² ; 0.1–1.5 V @ RT);	Li-foil	~475 (0.1 mA cm ⁻²)	ref. 180
CH ₃ NH ₃ PbIBr ₂ /ITO			~425 (0.1 mA cm ⁻²)	
CH ₃ NH ₃ PbI ₃ /ITO	1 M LiPF ₆ EC + DEC (1 : 1 vol ratio)		~340 (0.1 mA cm ⁻²)	
CsPbBr ₃ ; Conductive carbon black: PVDF binder (50 : 25 : 25 mass ratio) on copper foil	1 M LiTFSI in 1,3-dioxolane/dimethoxyethane (1 : 1 volume ratio)	Li-foil	102.6 (1 st cycle)	ref. 177
(MA) ₂ (PA) ₂ Pb ₃ Br ₁₀ ; AB: PVDF on Cu foil	Three-electrode cell (0.1 mA cm ⁻² ; 0.1–1.5 V @ RT); 1 M LiPF ₆ EC + DEC (1 : 1 vol ratio)	Li-foil	~375 (425 mA g ⁻¹)	ref. 180
C ₆ H ₉ I ₃ NOPb (1D); Conducting carbon: PVDF (60 : 30 : 10) on Cu foil	1 M LiPF ₆ in EC:DMC (1 : 1 ratio) + 5% FEC	Li foil	1580 (1 st cycle; 100 mA g ⁻¹)	ref. 179
(C ₄ H ₉ NH ₃) ₂ PbI ₄ (2D)			1605 (1 st cycle; 100 mA g ⁻¹)	
CH ₃ NH ₃ PbI ₃ (3D)			476 (1 st cycle; 100 mA g ⁻¹)	
C ₆ H ₉ I ₃ NOPb (1D); Conducting carbon: PVDF (60 : 30 : 10) on Cu foil	1 M NaClO ₄ in EC:DMC (1 : 1 ratio) + 5% FEC	Na-foil	961 (1 st cycle; 100 mA g ⁻¹)	
(EDBE)[CuCl ₄]; Conductive carbon black: PVDF (8 : 2 : 1 mass ratio) on copper or aluminum foils	1 M Li(TFSI) in dimethoxyethane	Li-foil	~25 (28 mA g ⁻¹)	ref. 172

The conversion reaction represented in eqn (1) involves phase separation (decomposition) of the perovskite, with Li being intercalated in the PbX₂ layered structure. The conversion

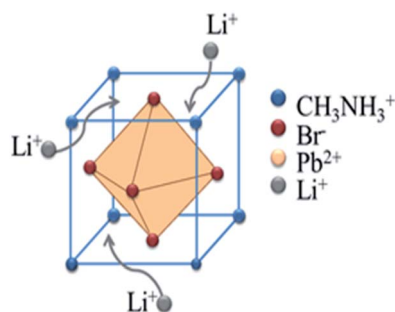


Fig. 13 Metal halide perovskite structure with B metals assembled around X anions to form a BX₆ octahedron. Cations fill the space formed by the eight corner-shared adjacent octahedra and balance the charge of the whole network. Multiple Li⁺ ion intake in the perovskite lattice is schematically represented. Reproduced from ref. 176 with permission (Copyright 2017, American Chemical Society).

reaction in eqn (2) is similar to the reaction found in conventional battery materials. The cation of a metal oxide or fluoride is reduced by Li metal to form the corresponding metal and lithium oxide/fluoride (*e.g.*, CoO + 2Li → Co + Li₂O).^{178,184,185} In the case of MHPs, the conversion reaction involves the formation of CH₃NH₃X, 2LiX, and metallic-Pb as products. Ramirez *et al.*¹⁸⁰ investigated the electrochemical properties of CH₃NH₃PbI_xBr_{3-x} with *x* = 0, 1, 2 in a simple three-electrode cell using MHP deposited films (500 nm; ~1 mg) on indium-tin oxide (ITO) as the working electrode and Li-foil as both counter and reference electrodes (Table 1). The Li extraction reaction achieved charge capacities of 70–160 mA h g⁻¹ for the first cycle, followed by an irreversible capacity. The calculated number of Li-ions per mole of electrode material was 9.7, 6.6, and 8.7 for CH₃NH₃PbI₃, CH₃NH₃PbIBr₂, and CH₃NH₃PbBr₃, respectively. Improvement in charge–discharge capacity was found when iodine was completely substituted by bromine. The MHPs with partial substitution (CH₃NH₃PbIBr₂) led to lower charge–discharge capacity, indicating a competition between the crystalline structure and bromine content. A recent study on

Li-ion batteries assembled with the CsPbBr₃ composite as the working electrode achieved the first charge–discharge capacity of 94.8 and 102.6 mA h g^{−1}, respectively.¹⁷⁷ XRD patterns of CsPbBr₃ before and after electrochemical lithiation showed a slight increase of the lattice constant confirming Li intercalation. Furthermore, metallic-Pb was probed by X-ray photoelectron spectroscopy (XPS) on CsPbBr₃ after the electrochemical lithiation using a wider potential window that also relates to eqn (2).¹⁷⁷ These initial reports show that the composition of 3D ABX₃ MHPs plays a major role in the Li-ion storage performance. Considering the structural diversity, better performance of Li-ion storage may be achieved by tuning the 3D MHP composition.

One of the key aspects of material properties for commercialization in Li ion battery technology concerns the high tolerance for volume expansion (*i.e.*, preservation of structural integrity) over many charge–discharge cycles and high reversibility with minimum instabilities during the phase transition of active materials and uncontrolled reactions at the electrode/electrolyte interfaces.^{182,186} Among the different structures of MHPs, low-dimensional MHPs are generally reported to be more robust in terms of long-term stability in performance over time as reported in several studies in the field of photovoltaics.^{187–190} In the field of Li ion batteries, low-dimensional MHPs have been demonstrated to further show superior

performance when employed as anodes for ion batteries because they can allow more intercalation into the free space between layers.^{173,179,180} Tathavadekar *et al.*¹⁷⁹ performed a systematic study on C₆H₉I₃NOPb (one dimensional, 1D), (C₄H₉NH₃)₂PbI₄ (two dimensional, 2D), and CH₃NH₃PbI₃ (3D) MHPs (Fig. 14) for accessing the specific capacity and stability associated with MHP dimensionality. The specific capacity (tested up 250 cycles) and stability improved dramatically by tuning the dimensionality of MHPs from 3D to 2D to 1D (Fig. 14g and Table 1). The 3D CH₃NH₃PbI₃ showed a first discharge capacity of 476 mA h g^{−1} at a current density of 100 mA g^{−1}. More dramatic enhanced performance data in the case of 2D and 1D with 1605 mA h g^{−1} and 1580 mA h g^{−1} were reported, respectively, which are nearly four times higher than that of the 3D case. These high first discharge capacities suggest that the low dimensional structures provided better access for Li-ions to the lead atoms in the MHP structures with which Li_xPb alloys are formed. The large drops from the first discharge capacity to the first reversible capacity for all the 3D–2D–1D cases are attributed to the irreversible reactions in the electrode material and solid electrolyte interphase (SEI) formation. However, it has been proposed that the formation of a stable SEI layer prevents further degradation of the MHP material observed from a stable capacity in subsequent cycles. As shown in Fig. 14h, the benzidine-based 1D case showed a robust

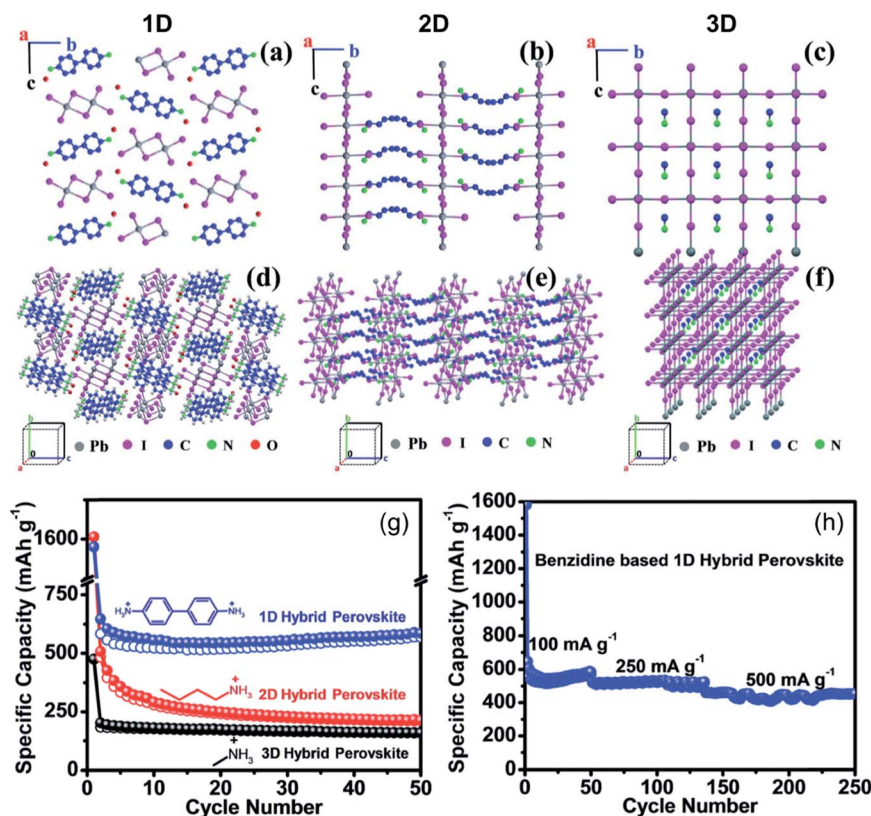


Fig. 14 (a) Perovskite structures of (a) C₆H₉I₃NOPb (1D), (b) (C₄H₉NH₃)₂PbI₄ (2D), and (c) CH₃NH₃PbI₃ (3D). (d), (e), and (f) represent the corresponding 3D perspectives. (g) Galvanostatic charge–discharge cycling stability measurements for 1–2–3D perovskites at a current density of 100 mA g^{−1} in the potential window of 2.5 to 0.01 V. (h) Cycling stability at different charge–discharge rates varying from 100 mA g^{−1} to 500 mA g^{−1} in the potential window of 2.5 to 0.01 V for the 1D perovskite case up to 250 cycles. Reproduced from ref. 179 with permission (Copyright 2017, The Royal Society of Chemistry).

cycling performance holding the capacity at $\sim 585 \text{ mA h g}^{-1}$ at a current density of 100 mA g^{-1} up to 50 cycles, $\sim 525 \text{ mA h g}^{-1}$ at a current density of 250 mA g^{-1} up to 136 cycles, and $\sim 453 \text{ mA h g}^{-1}$ at a current density of 500 mA g^{-1} up to 250 cycles. Ramirez *et al.*¹⁸⁰ have also studied the performance of another 2D class of semiconducting MHPs (Ruddlesden-Popper (RP) layered perovskites), namely propylammonium-methylammonium lead bromide $(\text{CH}_3\text{NH}_3)_2(\text{CH}_3(\text{CH}_2)_2\text{NH}_3)_2\text{Pb}_3\text{Br}_{10}$ consisting of an alternating arrangement of large organic layers and inorganic layers. The layered structure in MHPs was demonstrated to be important for allowing reversibility of the Li insertion-extraction of 100% with capacities of $\sim 375 \text{ mA h g}^{-1}$. On the basis of electrochemical impedance spectroscopy (EIS), cyclic voltammogram (CV), charge-discharge profiles, and XRD, the proposed mechanism for lithiation consists of a Li-ion intercalation mechanism in the MHP material followed by the $\text{Pb(II)} \rightarrow \text{Pb(0)}$ reduction process and alloying-dealloying process of the Li_xPb intermetallic compounds (LiPb , Li_8Pb_3 , $\text{Li}_{10}\text{Pb}_3$, and $\text{Li}_{22}\text{Pb}_5$).^{179,180} However, in the delithiation process the $\text{Pb(0)} \rightarrow \text{Pb(II)}$ reaction is not favorable, and as a result Pb-metal centers remain in the matrix of organic molecules. This MHP disintegration is not harmful because the matrix of organic molecules functions as a strain absorbing environment (cushion effect) against the typical electrode pulverization due to volume expansion. Furthermore, the presence of organic parts in MHPs was suggested to help avoid the detachment of Li_xPb alloys from the electrode.¹⁷⁹

The toxicity of a material applied to technology is another concern for investors and consumers, where handling mistakes during fabrication, leakage during operation, or the disposal phase may cause severe damage to the environment and to human beings.^{191,192} Initial studies have attempted the use of 2D MHP materials composed of a conducting organic material and Pb-free active inorganic material as Li-ion storage materials.^{171,172,175} However, the reported performances are low with typical specific capacity less than 50 mA h g^{-1} (Table 1).^{171,172,175} It has been proposed that Sn and Cu-based systems may be viable options as replacements for the Pb-case; for example, Sn was calculated to have higher alloying capacity than Pb.^{172,179} The MHP research applied to energy storage technology is still at a premature stage and further concerted efforts are needed to claim any superiority of these materials over previously researched non-perovskite based anode materials. However, based on our survey of reported studies, the concept of low-dimensional materials may have important implications for further improvements of MHP-based electrodes as well as development of new useful materials for energy storage applications.

5.2 Photo-rechargeable batteries

Batteries are energy limited and require recharging. On the other hand, batteries can be used to address the intermittency concern of solar cells. Development of photovoltaic-battery integrated systems with high energy density, efficiency, stability, and low-cost is a topic of importance for uninterrupted utilization of electricity generated from solar energy.

Great efforts are being made in this direction with several proposed innovative designs.^{12,193–195} Two design principles are often discussed: (i) conventional design of solar charging batteries that involves use of batteries and solar modules as two separate units connected by electrical wires (Fig. 15a and b)^{193,194}; (ii) advanced design that involves the integration of *in situ* battery storage in solar modules and therefore, offering compactness and fewer packaging requirements with the potential of low-cost (Fig. 15c).¹² The overall efficiency of an integrated solar cell-battery system is a product of sun energy-to-electricity power conversion efficiency (PCE) of the solar cell and energy storage efficiency of the battery. An insightful review article by Gurung *et al.*¹⁹⁵ describes the current status on solar charging battery systems. In this section, because of our research group expertise and concerted efforts towards Li-ion batteries and MHP based solar cells, we limit our description to the above two integrated systems focused on the use of MHPs.^{43,191,192,196–198} Xu *et al.*¹⁹³ demonstrated that four single junction $\text{CH}_3\text{NH}_3\text{PbI}_3$ based perovskite solar cells connected in series could be used to directly photo-charge a Li-ion battery composed of an LiFePO_4 cathode and an $\text{Li}_4\text{Ti}_5\text{O}_{12}$ anode (Fig. 15a). This system achieved an overall photoelectric conversion and storage efficiency of 7.8% at 0.1C and 7.36% at 0.5C discharge. This integrated device showed an energy storage efficiency of $\sim 60\%$. Gurung *et al.*¹⁹⁴ performed a similar study, but they used a single $\text{CH}_3\text{NH}_3\text{PbI}_3$ based perovskite solar cell adding inline a commercially available low-power direct current-direct current (DC-DC) boost converter (BQ25504EVM; Texas Instruments) with maximum power point tracking (MPPT) to efficiently charge their integrated Li-ion battery composed of an LiCoO_2 cathode and an $\text{Li}_4\text{Ti}_5\text{O}_{12}$ anode (Fig. 15b). The DC-DC converter circuit helps boost the low voltage of the single junction solar cell to the required charging voltage of 2.4 V Li-ion battery. In addition, this DC-DC converter circuit extracts the maximum power from the perovskite solar cell by the MPPT. This approach leads to the highest reported overall efficiency of 9.36% and average storage efficiency of 77.2% at 0.5C discharge. This work shows that it is important to have an appropriate matching of voltage and current between solar cells and batteries for attaining high efficiency by means of an external MPP tracker. Alternatively, a simpler design of a single device with merged functionalities of sun energy harvesting and storage was reported by Ahmad *et al.*¹² (Fig. 15c). 2D perovskite materials $[(\text{RNH}_3)_2\text{MX}_4]$ with R = organic moiety, M = metal, X = halide can be used for both energy storage (battery functionality) and photo-charging (photovoltaic functionality) simultaneously. The authors studied first the application of 2D lead halide based perovskites, $(\text{C}_6\text{H}_9\text{C}_2\text{H}_4\text{NH}_3)_2\text{PbI}_4$ (CHPI) and $(\text{C}_6\text{H}_9\text{C}_2\text{H}_4\text{NH}_3)_2\text{PbBr}_4$ (CHPB) as anodes in coin cell nonphoto-chargeable batteries. Capacities of $\sim 90\text{--}100 \text{ mA h g}^{-1}$ and $\sim 410 \text{ mA h g}^{-1}$ were achieved in the first cycle employing CHPI and CHPB, respectively. The hybrid solar cell-battery device architecture is composed of a 2D perovskite-rGO-PVDF (rGO = reduced graphene oxide; PVDF = poly(vinylidene fluoride)) film sandwiched between a separator and a transparent collector electrode fluorine-doped tin oxide, FTO. This photo-battery is first

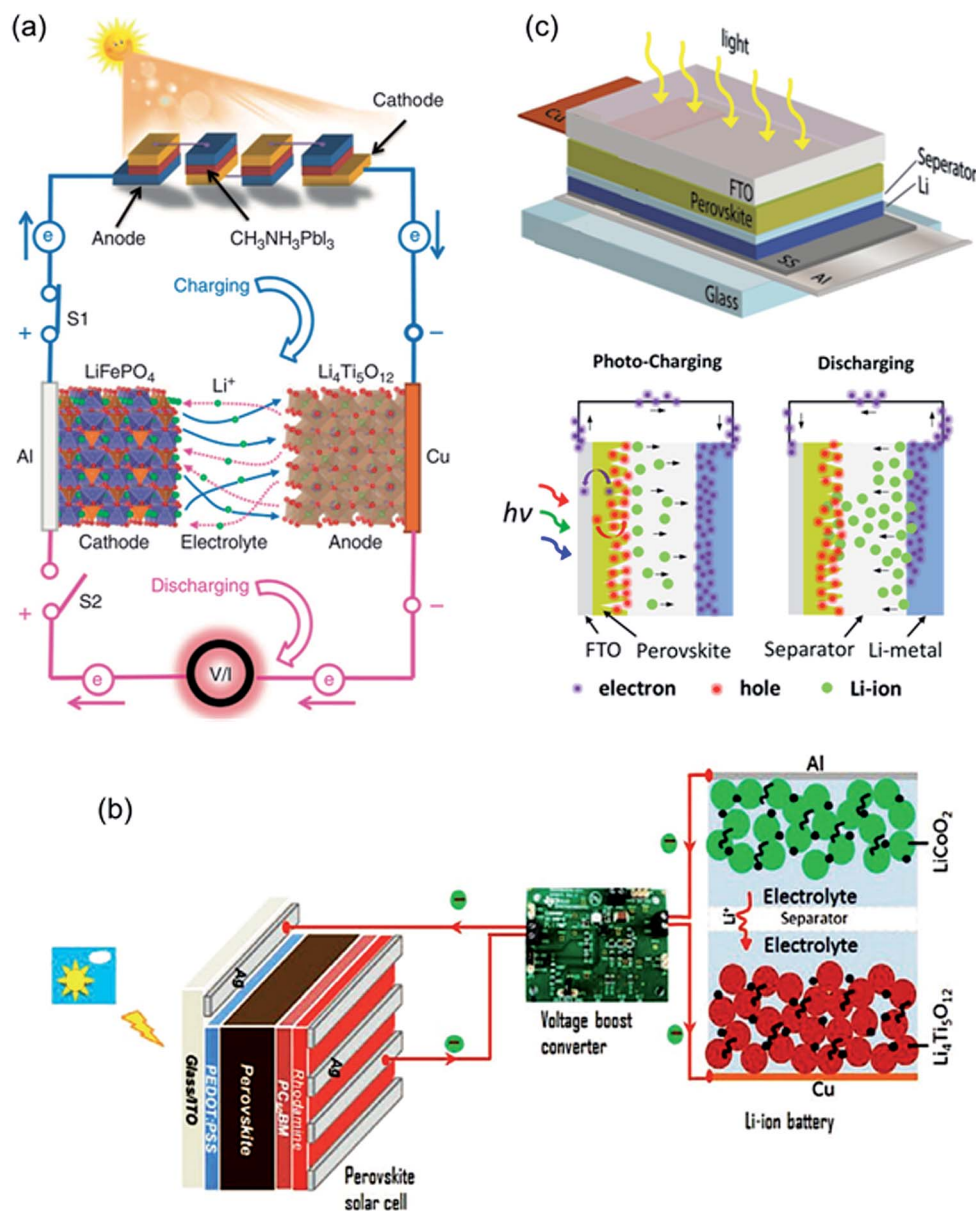


Fig. 15 (a) Photo-charging of a Li ion battery by a perovskite solar cell. Reproduced from ref. 193 with permission (Copyright 2015, Nature Publishing Group). (b) Photo-charging diagram of an $\text{Li}_4\text{Ti}_5\text{O}_{12}$ – LiCoO_2 cell using a $\text{CH}_3\text{NH}_3\text{PbI}_3$ based perovskite solar cell. The BQ25504EVM converter (Texas Instruments) is employed to provide DC to DC boost conversion and maximum power point tracking capability. Reproduced from ref. 194 with permission (Copyright 2017, WILEY-VCH). (c) Device structure and schematic representation of photo-charging and discharging of a perovskite photo-battery. Reproduced from ref. 12 with permission (Copyright 2018, American Chemical Society).

charged to saturation by a broadband light source ($\lambda \sim 420$ – 650 nm; 100 mW cm^{-2}) and then discharged through an LED. A capacity of up to 100 mA h g^{-1} can be delivered in this photo-battery. The proposed mechanism of energy storage is shown in Fig. 15c. Li ions migrate to the 2D perovskite anode where a combination of intercalation and conversion mechanism takes place.¹² Another new type of solid-state electrochromic batteries powered by perovskite solar cells have been designed by Xia *et al.*¹⁹⁹ These electrochromic batteries were demonstrated to deliver a capacity of 75 mA h g^{-1} at 1 A g^{-1} . Sustainability and clean energy have driven the innovation and development of advanced technologies of solar cell and energy

storage materials and technologies. The integrated solar cell–battery system technology is still at a very early stage, but for a few reports demonstrating the feasibility to harvest solar energy to charge batteries. In particular, the design aiming integration of *in situ* battery storage in solar cells is of great advantage offering compactness and fewer packaging requirements with the potential of low-cost. Commercialization of such integrated systems is still far from the present and it will depend significantly on the individual material and device architecture in both solar cell and battery research. Regarding this aspect, MHP materials may be key for the advancement of such technology. The exceptional performance of MHPs as sun

energy harvester materials for solar cells was demonstrated and ascribed to their excellent material properties such as direct band gap, outstanding high absorption coefficient, abrupt optical band edge, large charge carrier diffusion length and low exciton binding energy. The exploration of MHP materials in the field of Li-battery is still scarce. However, the initial studies have demonstrated that low-dimensional MHPs show superior performance when employed as anodes because they can allow more intercalation into the free space or within the organic layers.^{173,179,180} Furthermore, low-dimensional MHP materials provide key aspects of high tolerance for volume expansion over many charge–discharge cycles, high reversibility with minimum instabilities during phase transition in active materials, and controlled reactions at the electrode/electrolyte interfaces.^{182,186} The ABX₃ chemical composition engineering and/or alloying strategy has been scarcely explored and further exploration for an optimal composition is expected to bring subsequent years of key findings for the Li-ion battery community and solar cell–battery integrated systems.

6. Conclusion and outlook

The evolution of lithium-ion batteries strongly relies on the development of working electrodes. Notwithstanding that the current review covers a limited number of electrode types, the surveys certainly add to the comprehensive outlook on the advances of the battery technologies. In the early period, different classes of electrode materials were proposed such as olivine, spinel, and layered materials. This classification is based on the type of Li⁺ (ionic) diffusion pathways rather than the chemical composition. Since the successful commercialization of LCO, developing batteries with a balance between discharge capacity, thermal stability, capacity retention, and lifespan has become a central issue of battery research. This leads to the development of mixed transition metal oxide cathodes (*e.g.* NMC), which can provide synergetic benefits over a single transition metal cathode. However, growing demands for lithium-ion batteries to operate with higher discharge capacities and longer cycle numbers have led to a renewed interest in tailoring the existing NMC electrode. The development of Li-rich (or excess) layered oxide cathodes, often denoted as $x\text{Li}_2\text{MnO} \cdot (1-x)\text{LiTO}_2$ ($T = \text{Mn, Ni, Co}$), has drawn much attention due to their superior theoretical capacity ($>250 \text{ mA h g}^{-1}$) with high operation voltage ($>4.6 \text{ V vs. Li}^0$). As discussed, the origin of the high charge capacity in this battery system can be explained by the intimate integration between the (001) planes from Li_2MnO_3 and the (003) planes from LiTO_2 with the interlayer spacing of $\sim 4.7 \text{ \AA}$ by the interleaving of close-packed layers. Simply, this can be regarded as an NMC cathode with a Li⁺ booster by Li_2MnO_3 . The development of Li-rich transition metal oxide cathodes involves both compositional and structural modifications of the conventional material. The ever-growing markets of electric vehicles (EVs or HEVs) are now intensively pursuing power sources with higher energy densities. In this regard, Ni-rich layered oxide batteries are likely to be a promising solution to meet the urgent demand, due to their high discharge capacity ($\sim 220 \text{ mA h g}^{-1}$). The two-stage

oxidation changes between $\text{Ni}^{2+}/\text{Ni}^{3+}$ and $\text{Ni}^{3+}/\text{Ni}^{4+}$ significantly contribute to the charge compensation mechanism, and thus extra capacities can be achieved when the NMC cathode has a higher Ni concentration. Although the high-capacity feature is derived from the conventional LiNiO_2 , recent studies have attempted to find the precise and optimum stoichiometric ratio between Ni (high concentration), Mn, and Co to develop a high-performance battery with improved stability and/or good rate capability. We therefore consider the emergence of Li-rich and Ni-rich layered oxides as the initial evolution towards advanced technologies for high energy and power-density lithium-ion batteries. However, there are also critical drawbacks of these battery systems. Li-rich layered oxide batteries undergo an irreversible capacity loss during the initial cycles. At a higher cut-off voltage ($>4.5 \text{ V}$), lithia (Li_2O) is extracted from Li_2MnO_3 with phase transformations, resulting in dramatic capacity fading during subsequent charge and discharge. Also, Ni-rich layered oxide batteries suffer from electrode degradation by the undesired redox reactions with the electrolyte. Surface reconstruction or clustering by NiO at higher voltage ranges is another major problem which needs to be addressed. Engineering the active material by substituting with other transition metals or employing surface coatings is likely to be helpful in ameliorating the battery performance and stability. Moreover, replacing conventional electrolytes with specially designed electrolytes can be a reasonable treatment for mitigating the aggressive electrolyte decomposition at a high operating voltage. However, some inherent limitations persistently remain, preventing them from being used for high-capacity batteries. Hence, alternative material technology, *i.e.*, conversion electrodes, has been renewed in recent years. This old concept (chemistry) is rising again, leading to the development of new energy storage technologies such as Li–sulfur, Li–air, and Li–CO₂ batteries. We regard this as an evolution into the state-of-the-art technologies for lithium-ion batteries. Owing to the conversion mechanism, the discharge capacity and the energy-density could be dramatically enhanced. However, there are also technical barriers that limit the successful commercialization of these applications such as over-potential (Li–air and Li–CO₂), shuttling effect (Li–sulfur), poor rate capability, high interphase resistance (solid-state batteries) and low cycle life.

We are also anticipating the evolution of lithium-ion battery technologies towards self-sufficiency, namely, a stand-alone energy system (or off-grid) for autonomous electronic devices has drawn much attention, recently. The major advantage of using this technology is the high energy efficiency with reliable power at reasonable cost, providing electricity in a remote location. Photo-rechargeable batteries are a good example of the stand-alone energy system, as they combine energy harvesting materials with energy storage materials. A key factor that leads to the successful development of the autonomous device is the seamless integration of the power source (*e.g.* battery electrode) into the application. Organohalide perovskite (*e.g.* $\text{CH}_3\text{NH}_3\text{-PbBr}_3$ and $\text{CH}_3\text{NH}_3\text{PbI}_3$) has emerged as a promising photo-voltaic material for solar-cells. A problem can arise, however, when an attempt is made to combine the organohalide

perovskite with the battery component without any considerations of the energy level alignment involving the conduction band, valence band, Fermi level, and work function between the electrodes. Successful hybridization with lithium-ion batteries relies on the good compatibility between the active materials.

The present review has shown key evolutions that push the lithium-ion battery technologies forward. Research will continue to push the limits of energy density, power density, rate capability, stability, safety, and cost. On the other hand, the direct incorporation of the energy storage component for the autonomous device would be a new trend of the battery technologies. We hope that this review can offer key insights into the development of the next-generation lithium-ion batteries.

Conflicts of interest

There are no conflicts to declare.

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